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# Sensor Networks and IoT: Architectures, Protocols, and Security

**Walid Bechkit,** TC@INSA-Lyon, Agora@INRIA

citi lab

Inria INVENTEURS DU MONDE NUMÉRIQUE

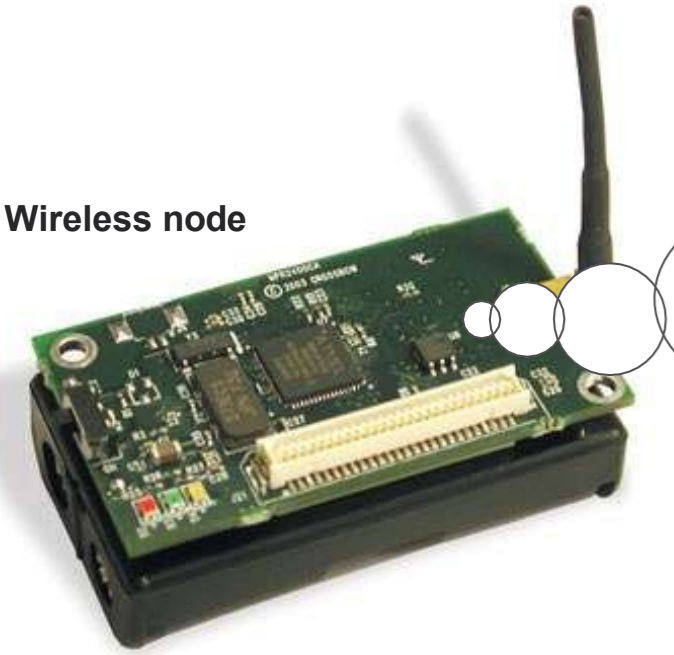
membre de UNIVERSITÉ DE LYON

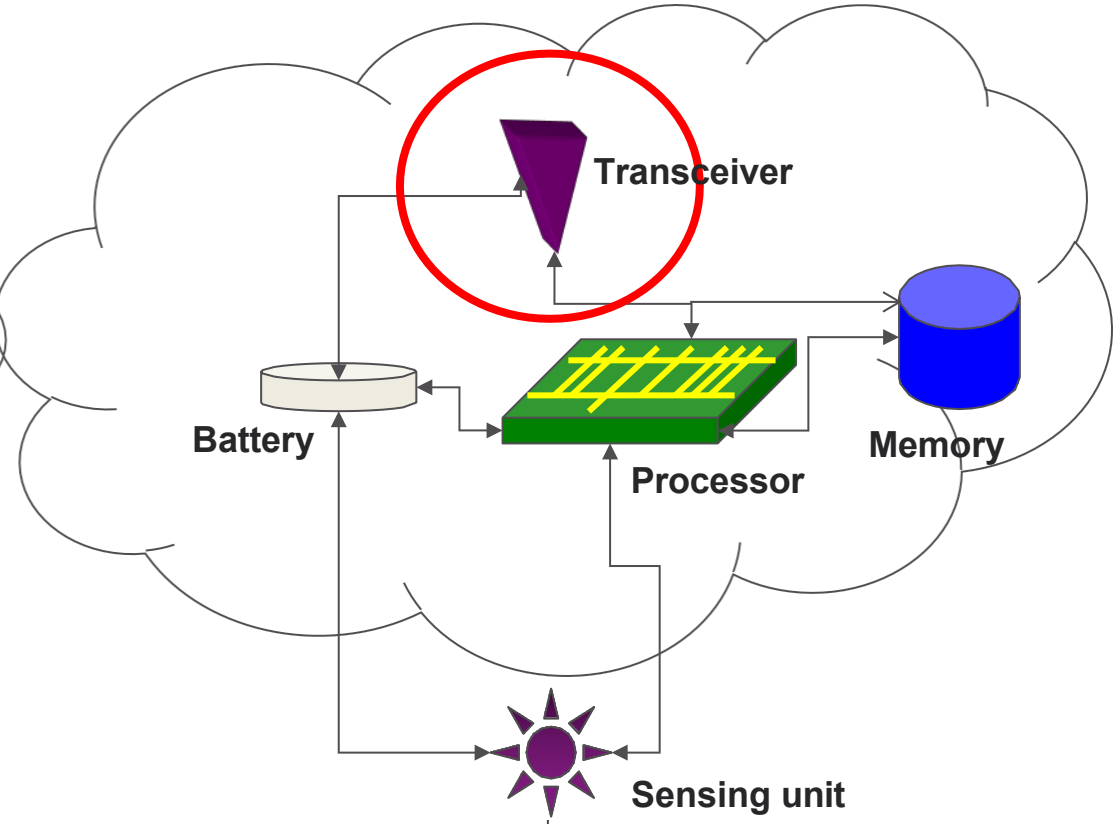
# 1.1 Internet of Things and Wireless Sensor Networks

## General context

# Sensor node architecture

Wireless node






Example of external sensors



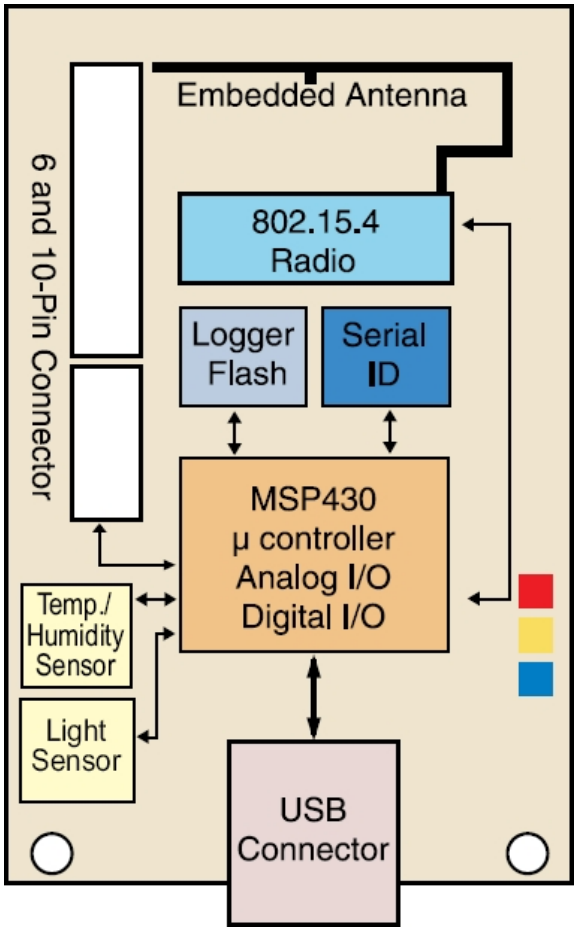
# Example of Telosb

- **Processor**
  - –TI MSP430
  - –8 MHz
  - –10kB RAM
- **Transmission**

IEEE 802.15.4
**(TI CC2420)**

  - 250 Kbps (2.4 - 2.4835 GHz Band)
  - Embedded antenna
- **Flash**
  - –1 MB
- **Sensors**

  Light, Temperature, and Humidity



# MKR 1010 board

- Power supply (micro-USB port, 3.7 LiPo battery, VIN pin = 5v)
- Microcontroller: SAMD21 Cortex M0+, 32-bit at 48 MHz
- Flash memory: 256 kB
- SRAM memory: 32 kB
- Output voltage: 3.3 Vdc

- **Secure element (ATECC508A)**
- 7 analog inputs ADC 8, 10 (default), and 12 bits
- 1 analog output DAC 10 bits

- 8 digital I/O
- SPI (MOSI, SCK, MISO)
- I2C (SCL, SDA)
- UART (TX, RX)

+ WiFi & Bluetooth
(u-blox NINA-W102)



# MKR 1310 board

- Power supply (micro-USB port, 3.7 LiPo battery, VIN pin = 5v)
- Microcontroller: SAMD21 Cortex M0+, 32-bit at 48 MHz
- Flash memory: 256 kB
- SRAM memory: 32 kB
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- 7 analog inputs ADC 8, 10 (default) and 12 bits
- 1 analog output DAC 10 bits

- 8 digital I/O
- SPI (MOSI, SCK, MISO)
- I2C (SCL, SDA)
- UART (TX, RX)

- + LoRa / LoRaWAN (Murata CMWX1ZZABZ)



# IoT applications

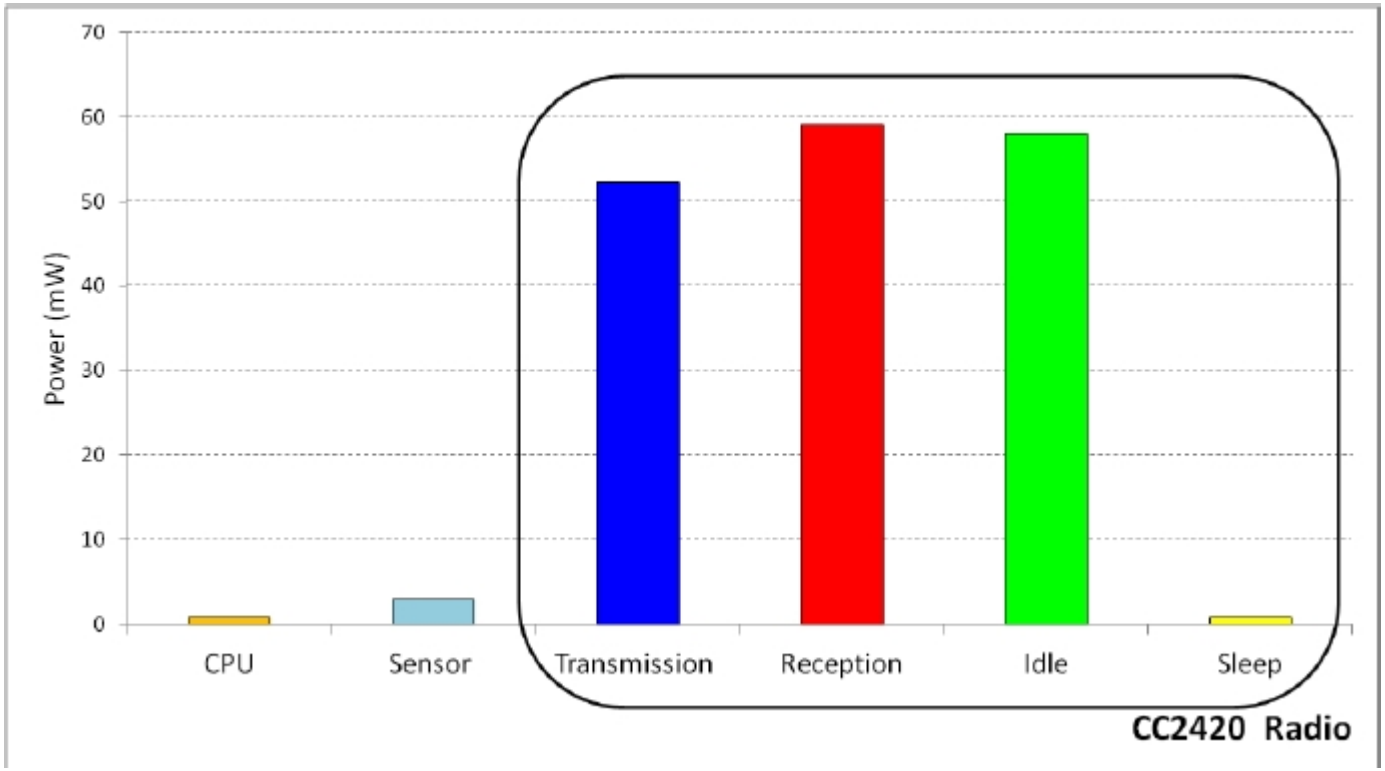


# Characteristics of IoT objects

- Resource limitations:
  - **Energy consumption constraints**
  - **Memory constraints**
- Lack of infrastructure
- Wireless environment
- Large-scale deployment
- Coupling with the physical environment
- Node mobility
- Node heterogeneity

# Energy constraints in sensor nodes

- Resource limitations, primarily energy
- Radio consumption is the most significant



# Security challenges in WSN

**IT life cycle**

- Factorized software
- Upgrades
- Soft remains when unusable
- Large scale
- Mature codebase

**IoT life cycle**

- Fragmentation
- No upgrades
- Non-soft restes
- Very large scale
- Young codebase

How IoT revisits security problems, François Lesueur, INSA-Lyon, April 2017

+ It's not the walls that protect the citadel, but the spirit of its inhabitants."

[Thucydides]

At least 40% of attacks are caused by users themselves (e.g., Mirai botnet).

# Security challenges in WSN

## Information Security

$Risk = Threat \times Vulnerability \times Cost$

### Vulnerability

- Deployment environment
- Lack of physical protection
- Memory limitation
- Energy limitation
- Wireless Communication
- Lack of infrastructure

### Threat

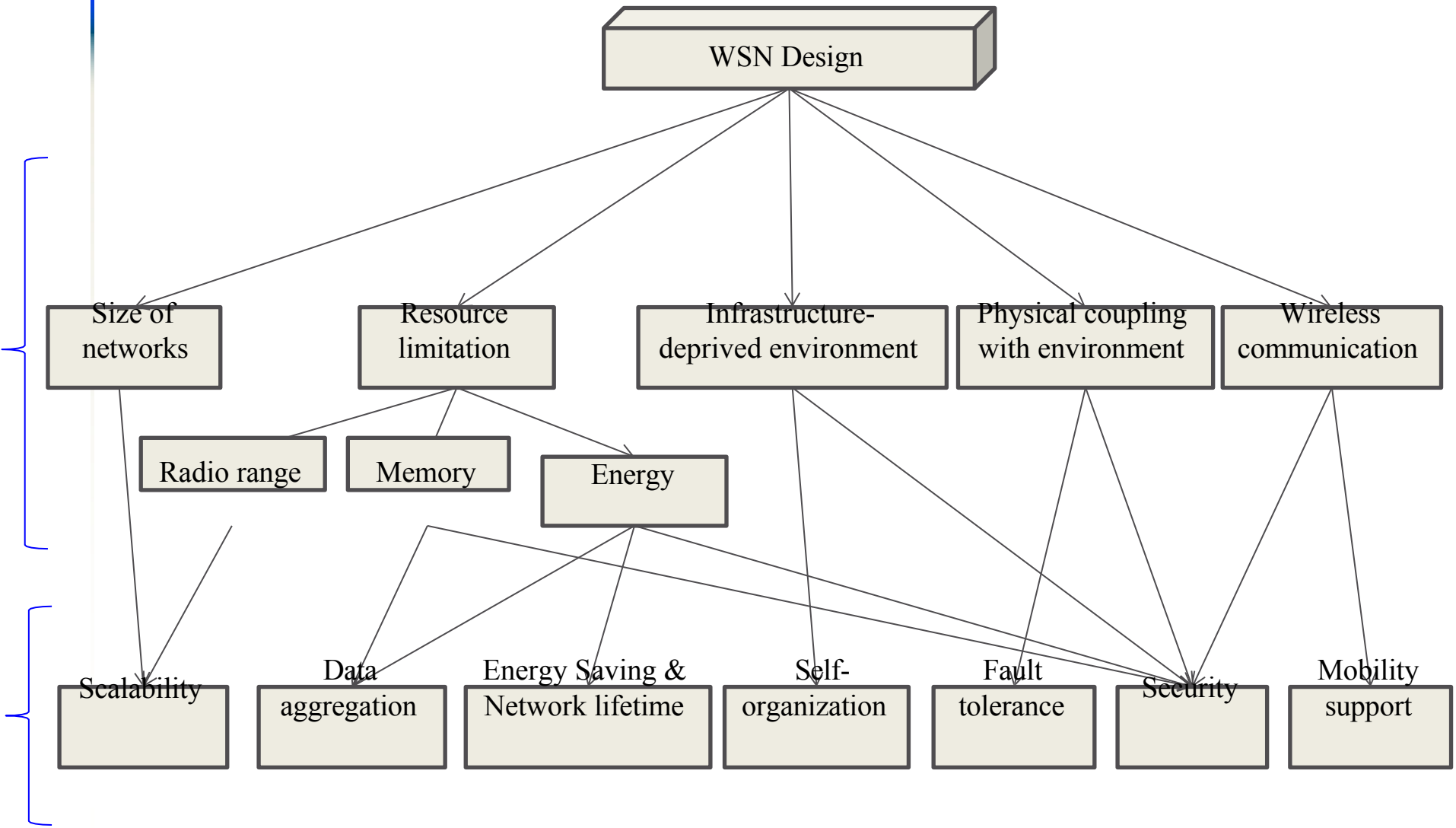
- Traditional threats (eavesdropping, message injection ...)
- New specific attacks (energy drain, data aggregation attacks)

### Cost

- Critical applications of WSN
- Dramatic effects and costs

Ex: Mirai botnet (DDoS attacks in 2016/17)

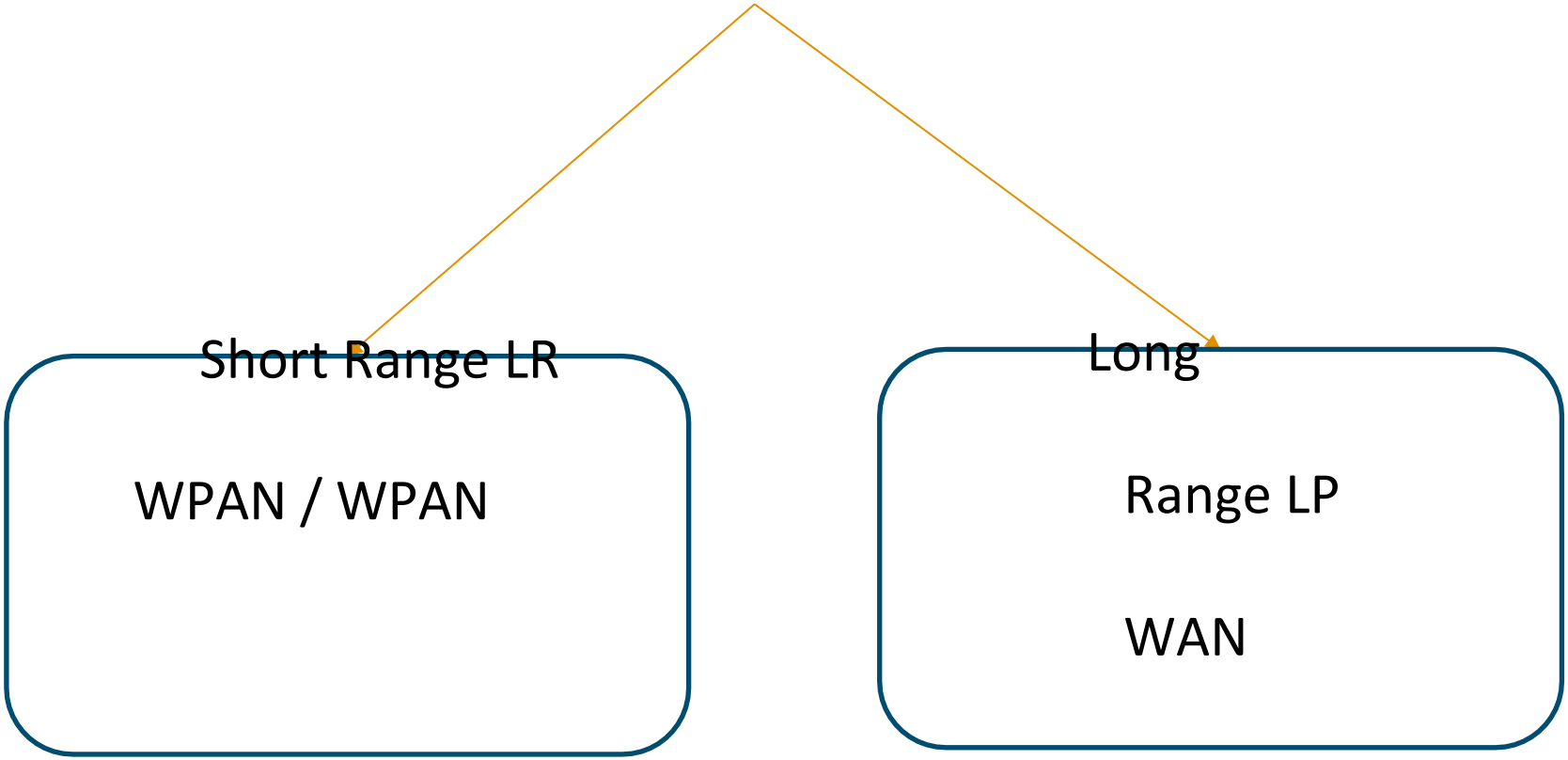
# WSN: From characteristics to challenges



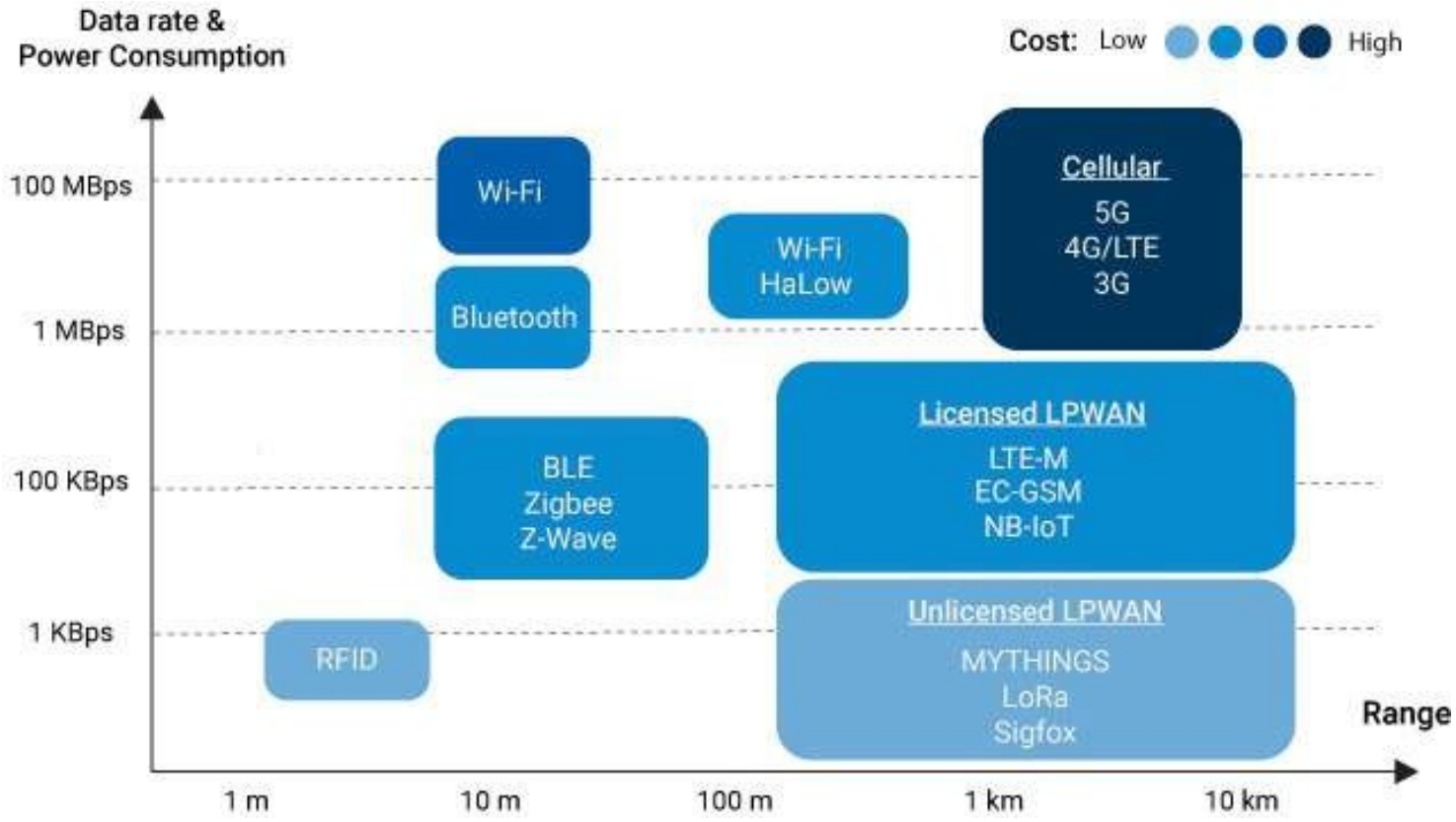
# 1.2 Low-energy consumption communication protocols

## Standardized solutions

# Low-energy wireless protocols (for IoT)

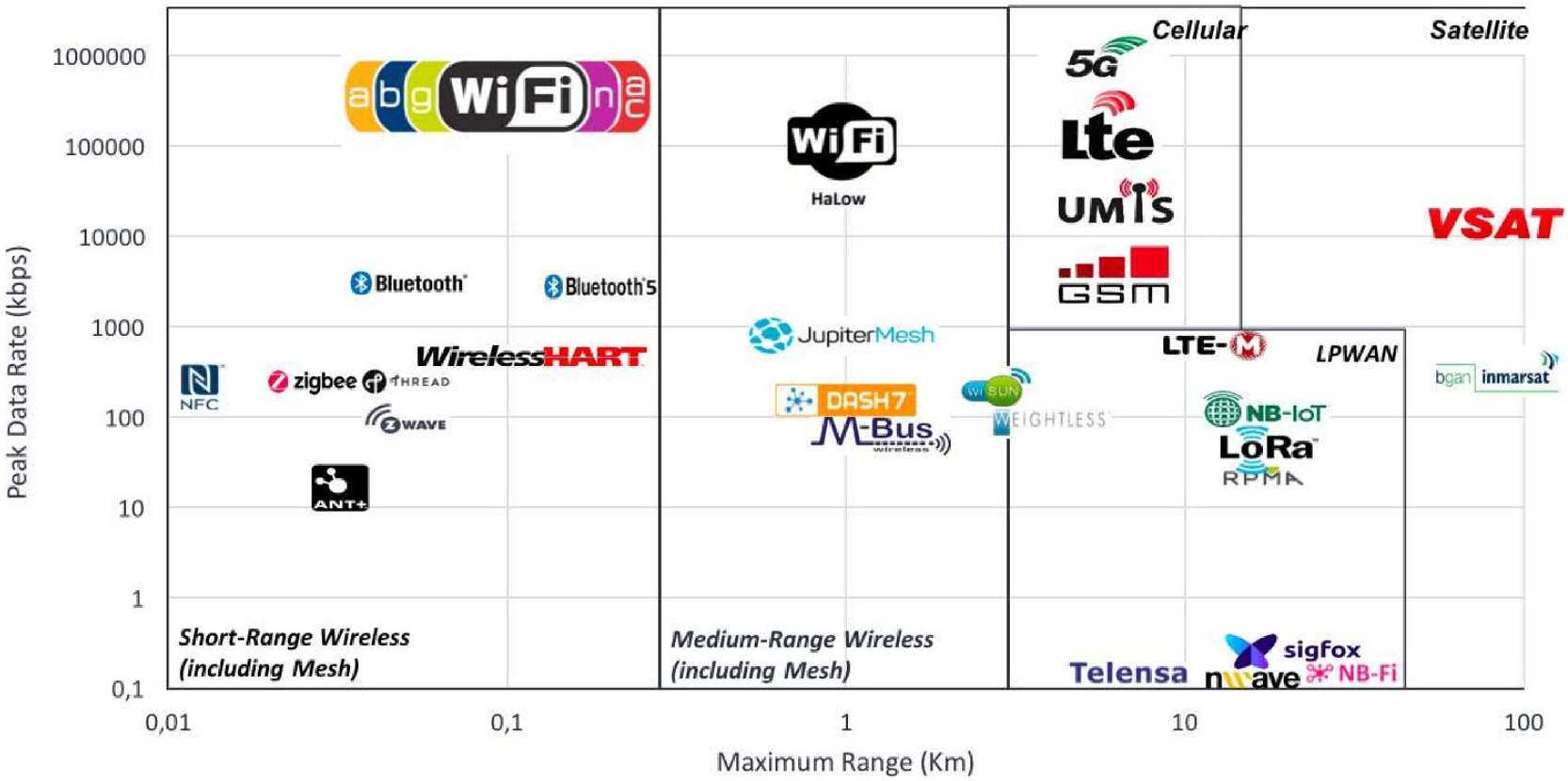


# Classification of main standards




Source: https://behrtech.com/blog/6-leading-types-of-iot-wireless-tech-and-their-best-use-cases/

# Classification of the main standards




Source: https://iot-analytics.com/iot-segments/iot-connectivity/

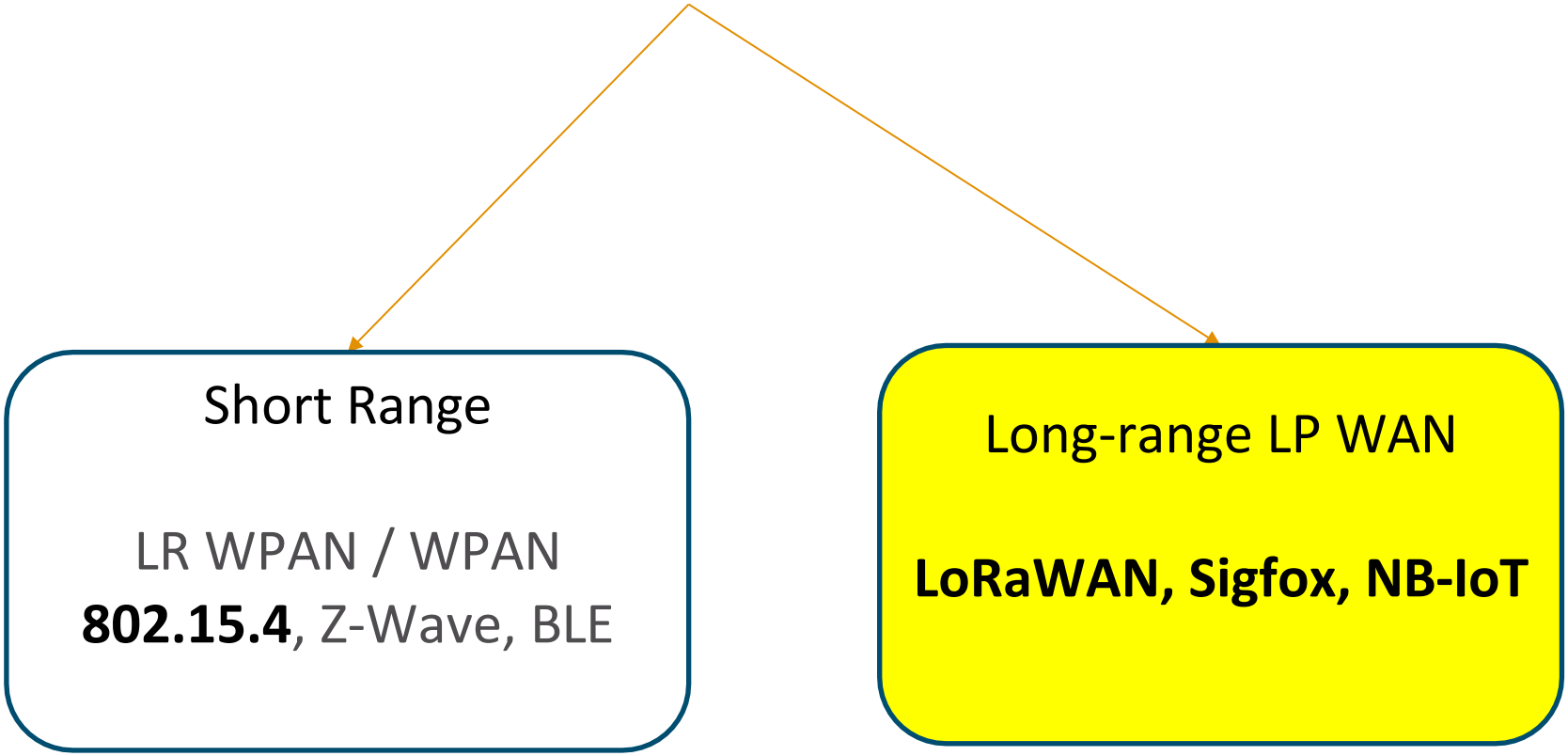
# Classification of the main standards

| Attribute | Bluetooth® Low Energy Technology | Wi-Fi | Z-Wave | IEEE 802.15.4 (Zigbee, Thread) | LTE-M | NB-IoT | Sigfox | LoRaWAN |
|---|---|---|---|---|---|---|---|---|
| Range | 10 m – 1.5 km | 15 m – 100 m | 30 m - 50 m | 30 m – 100 m | 1 km – 10 km | 1 km – 10 km | 3 km – 50 km | 2 km – 20 km |
| Throughput | 125 kbps – 2 Mbps | 54 Mbps – 1.3 Gbps | 10 kbps – 100 kbps | 20 kbps – 250 kbps | Up to 1 Mbps | Up to 200 kbps | Up to 100 bps | 10 kbps – 50 kbps |
| Power Consumption | Low | Medium | Low | Low | Medium | Low | Low | Low |
| Ongoing Cost | One-time | One-time | One-time | One-time | Recurring | Recurring | Recurring | One-time |
| Module Cost | Under $5 | Under $10 | Under $10 | $8-$15 | $8-$20 | $8-$20 | Under $5 | $8-$15 |
| Topology | P2P, Star, Mesh, Broadcast | Star, Mesh | Mesh | Mesh | Star | Star | Star | Star |
| Shipments in 2019 (millions) | ~3,500 | ~3,200 | ~120 | ~420 | ~7 | ~16 | ~10 | ~45 |

Source: https://www.bluetooth.com/blog/wireless-connectivity-options-for-iot-applications-technology-comparison/

# 1.3 Long-range solutions

# Wireless protocols for the IoT



# License-free frequencies: LoRa or Sigfox

sigfox




LoRa™

LoRaWAN™

# Long Range Low Power: SigFox / Lora

**Sigfox (2009**): supported by GDF Suez, Air Liquide, Telefónica, etc.)

**LoRa:** Cycléo 2009 acquired by Semtech **in 2012.**

**LoRa Alliance** Semtech Orange, Objenious by Bouygues Telecom, STMicroelectronics, Cisco, IBM, etc.

# Long Range Low Power: SigFox / Lora

| | SigFox | LoRa / LoRaWAN |
|---|---|---|
| ISM band | Yes | Yes |
| Subscription | Yes | Not necessarily |
| Bandwidth (kHz) | 192 kHz | 125-500 kHz |
| Modulation | UNB | CSS |
| Data rate (b/s) | 100 | ~ 250 – 11,000 (125 kHz) |
| Maximum range (km) | 63 | 22 |
| Maximum payload size (o) | 12 | 255 (Lora),<br>51 to 222 in LoRaWAN |
| Network type | Single operator | Private or operated network |
| Uplink | 140 messages/day | Unlimited (ISM duty cycle) |
| Downlink | 4 messages/day | Unlimited (ISM duty cycle) |

# LPWAN: summary of the situation in France

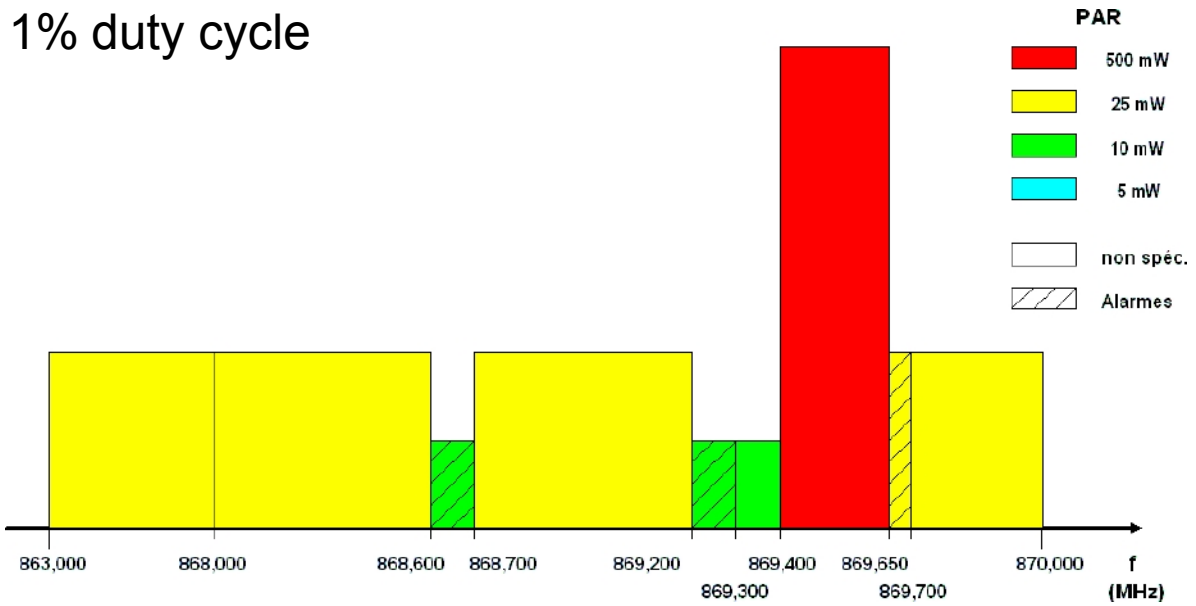
- Orange focuses on LoRa/LoRaWAN and LTE-M
- Bouygues via Objenious focuses on LoRa/LoRaWAN
- SFR focuses on NB-IoT
  - SigFox: global operator of SigFox networks
    → Acquired by UnaBiz in April 2022

# 1.3 Long-range solutions

## Example: LoRa/LoRaWAN

# LoRa: Fréquencies

- 433 or 868 MHz bands (in Europe), 915 MHz (in America), etc.
- In Europe: Effective Radiated Power limited to 10 mW across the entire 433 MHz band and variable across the 868 MHz band
- Mostly 1% duty cycle




Source of figure: http://www.onditech.com/documents/freq_433_868.pdf

# LoRa: Physical layer

**Spreading Factor (SF)**

**Transmission Power, Reception Sensitivity**

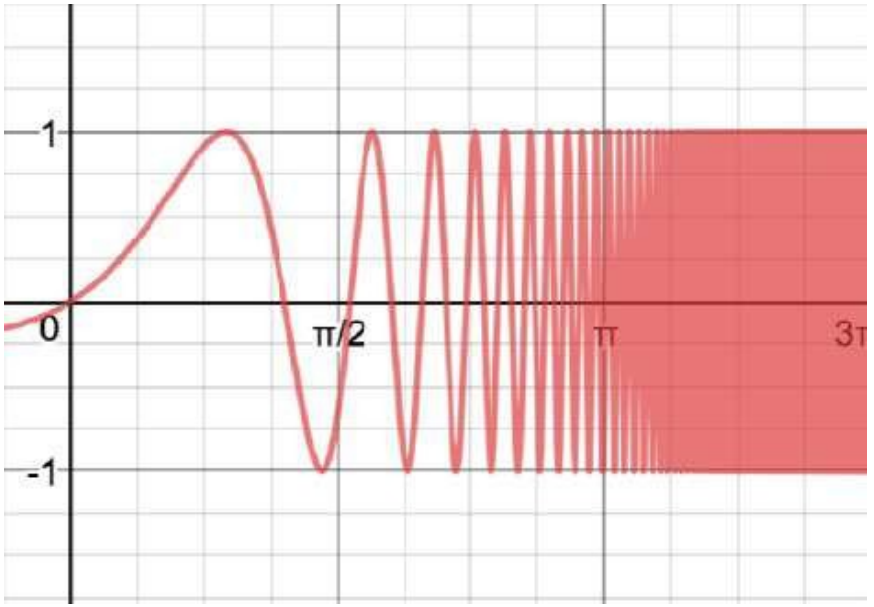
**Coding Rate (CR)**

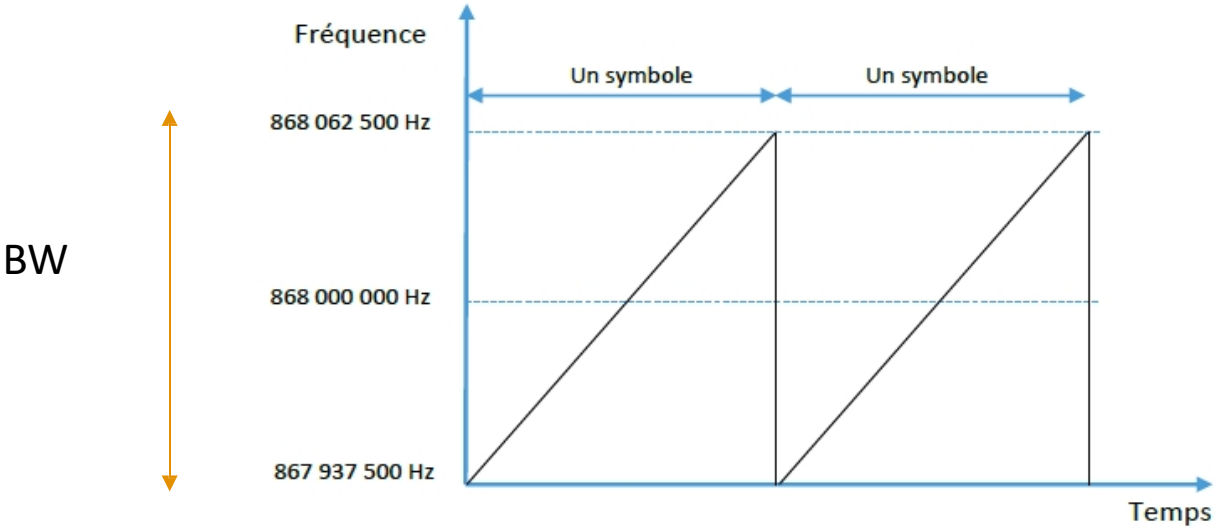
**Bandwidth**

# LoRa: Physical Layer (modulation)

- Modulation used

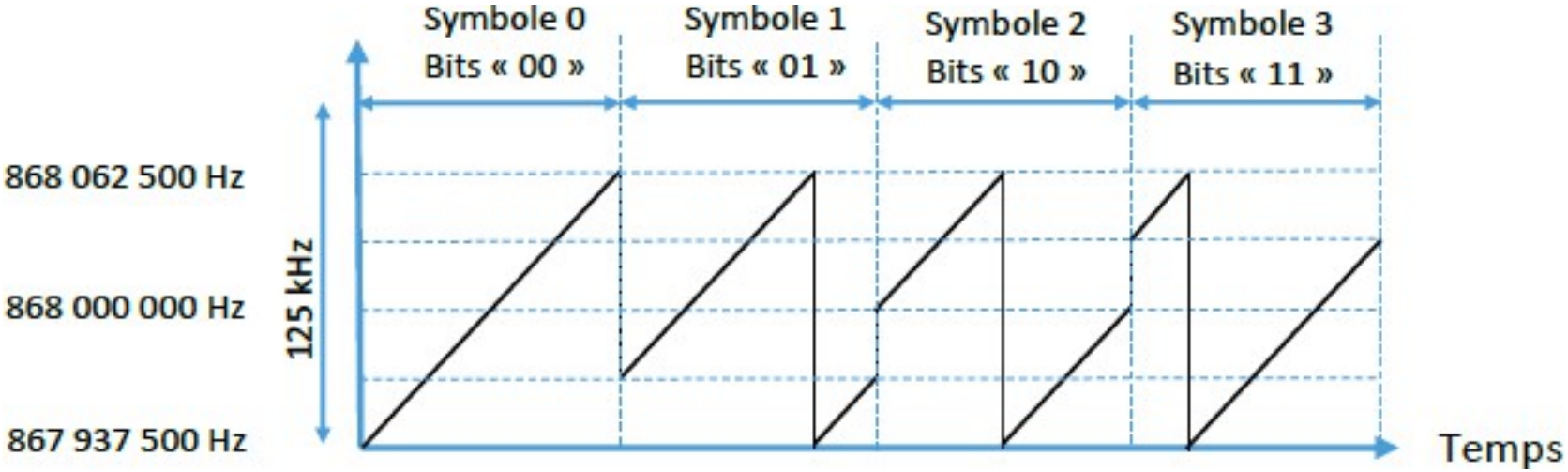
CSS (Chirp spread spectrum)



# LoRa: Physical Layer (SF and modulation)

- SF: number of bits per symbol  1 Symbol = SF bits
- Each symbol is represented by a chirp
- So we have $2^{SF}$ possible symbol shapes (chirp).
- Example: SF=2




Source: LoRa/LoRaWAN training course, Sylvain MONTAGNY, Savoie Mont Blanc University

# LoRa: Physical Layer (TX and Ps)

- **Maximum power in Europe 25 mW = 14 dBm**
- The great advantage of LoRa is that even very weak signals (down to -120 to -140 dBm) can be demodulated.
- It can be demodulated even with an SNR between -5dB and -20dB

| *SpreadingFactor* (RegModemConfig2) | Spreading Factor (Chips / symbol) | LoRa Demodulator SNR |
|---|---|---|
| 6 | 64 | -5 dB |
| 7 | 128 | -7.5 dB |
| 8 | 256 | -10 dB |
| 9 | 512 | -12.5 dB |
| 10 | 1024 | -15 dB |
| 11 | 2048 | -17.5 dB |
| 12 | 4096 | -20 dB |

Source: LoRa/LoRaWAN training course, Sylvain MONTAGNY, Savoie Mont Blanc University

# LoRa: Physical layer (bandwidth)

- Bandwidth
- The default values are generally 125 kHz, but 62.5 kHz, 250 kHz, 250 kHz, and 500 kHz are also used
- By doubling the bandwidth, the symbol time is halved and the data rate is doubled.

# LoRa: Physical Layer (C/R)

- Remember Forward Error Correction, redundancy at the physical level to detect and correct errors

- CR' = 4 / (4 + CR)
- CR = 1, 2, 3, or 4
- CR' = 4/5, 4/6, 4/7, or 4/8

| CR | CR’ | Surplus |
|---|---|---|
| 1 | 4/5 | 25 |
| 2 | 4/6 | 50% |
| 3 | 4/7 | 75% |
| 4 | 4/8 | 100% |

+ robust (error detection/correction)

More bits to send

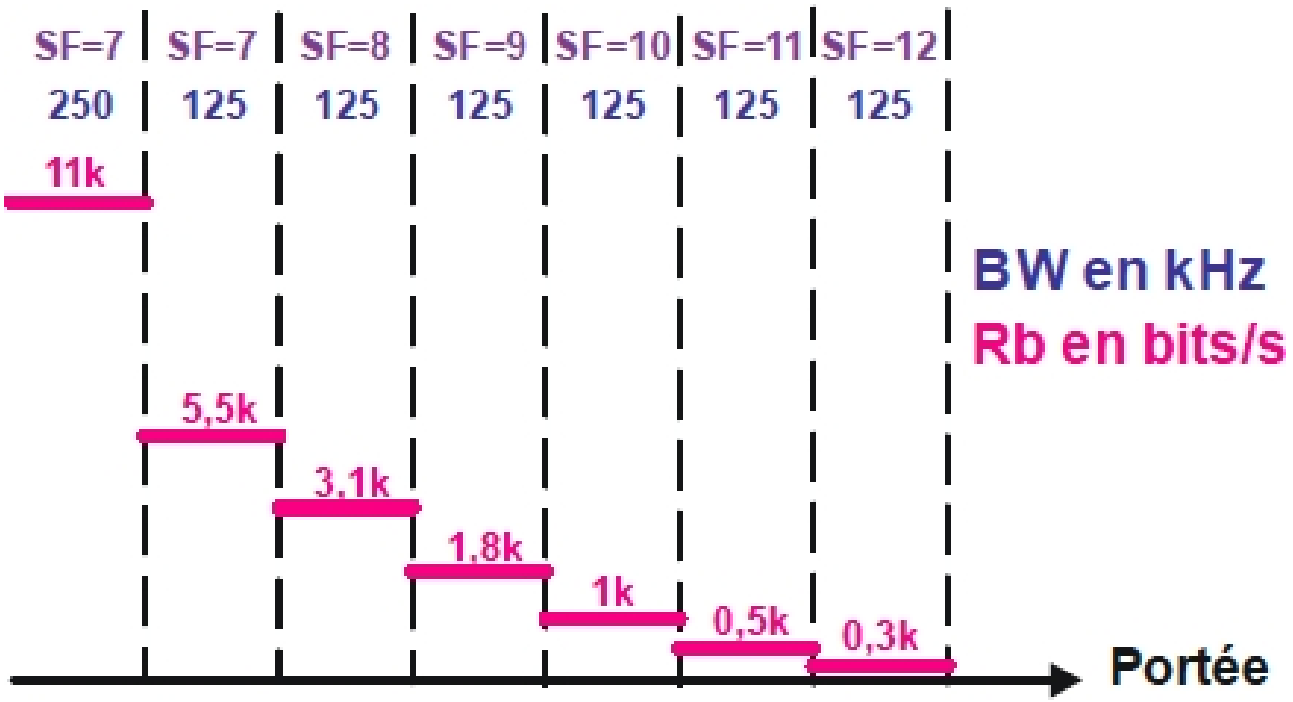
# LoRa: Physical Layer (Useful Data Rate)

**Theoretical throughput using the formula**

| SF | BW (KHz) | CR 4/5 | CR 4/8 |
|---|---|---|---|
| 7 | 250 | 10,938 | 6836 |
| 7 | 125 | 5469 | 3418 |
| 8 | 125 | 3125 | 1953 |
| 9 | 125 | 1758 | 1099 |
| 10 | 125 | 977 | 610 |
| 11 | 125 | 537 | 336 |
| 12 | 125 | 293 | 183 |

- Be careful with rounding
- The actual data rates are very similar (see TP ☺)

# LoRa: Physical layer (useful data rate CR4/5)




Source: Characterization of LoRa modulation and demodulation processes, J-F SERREAU, 2019

# LoRa: Summary

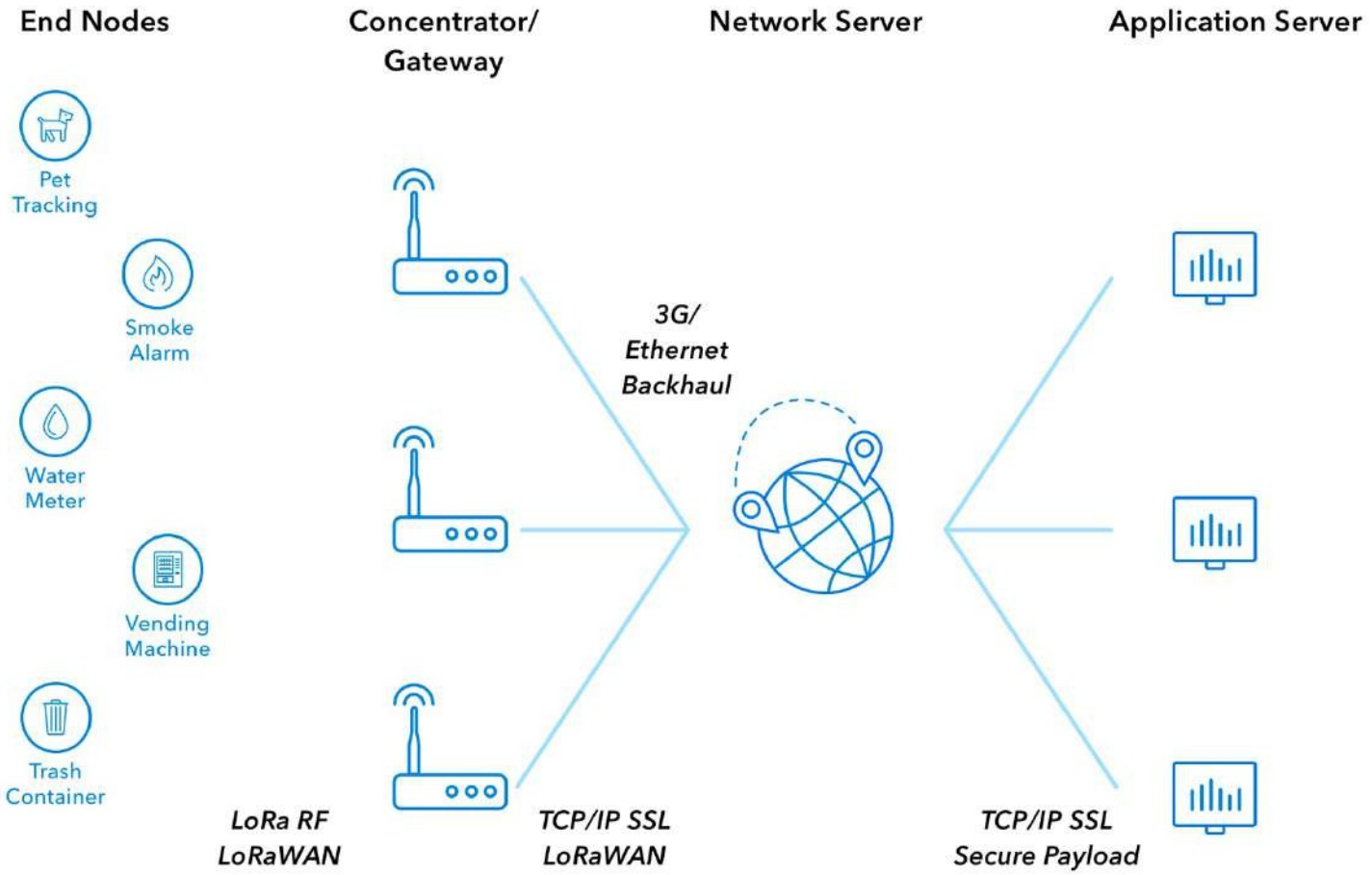
**Spreading Factor (SF)**

**Transmission power, Reception sensitivity**

**Coding Rate (CR)**

**Bandwidth**

# LoRaWAN: Architecture




**source:** https://www.thethingsnetwork.org/

# LoRaWAN: Architecture

- **Node (LoRaWAN device):** transmission of data (often from sensors) and reception of data in downlink mode.

- **Gateway (LoRaWAN Gateway):**
  - bridge between nodes and the core network (network and application services).
  - Listens on multiple LoRa channels for all SFs at once

- **Network server**:
  - Data integrity, message deduplication
  - Link to the application server, choice of gateway for downstream traffic.

- **Application server:**
  - Decrypting messages and presenting them to users

# LoRaWAN: Identifiers and keys to know

- **Identifiers to know:**
  - DevEUI - 64-bit physical identifier of the node (device): globally unique EUI-64, assigned at manufacture
  - DevAddr - 32-bit node address: dynamic, non-unique, assigned through the activation procedure (details in the next lesson)
  - GwEUI - 64-bit physical identifier of the gateway: EUI-64 (unique)
  - AppEUI - 64-bit application identifier: EUI-64, unique per network

- **Key points to remember (details in next class):**
  - AppKey: primary key used to derive session keys
  - AppSKey: session key for confidentiality (data encryption)
  - NwkSKey: data integrity session key.

# LoRaWAN: deployment of LoRaWAN resources

- **Operated networks**
  - Gateways, network servers, and application servers managed by operators. Examples in France: Orange and Objenious by Bouygues
- **Private networks**
  - Several open source network and application server solutions available
- **Community networks**
  - Example of TTN (community deployment of gateways), network servers, and application servers managed by TTN.

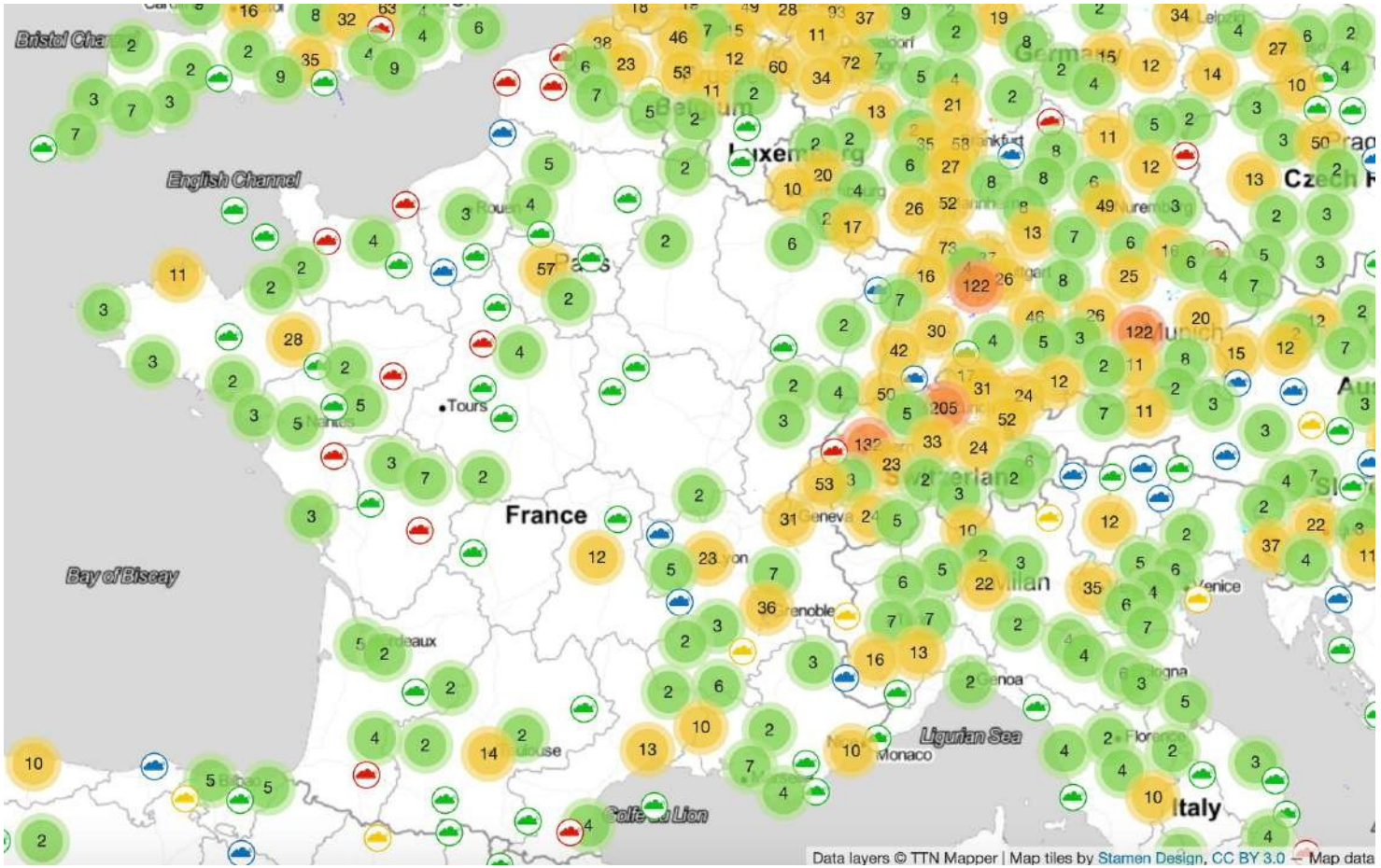
# LoRaWAN: Coverage



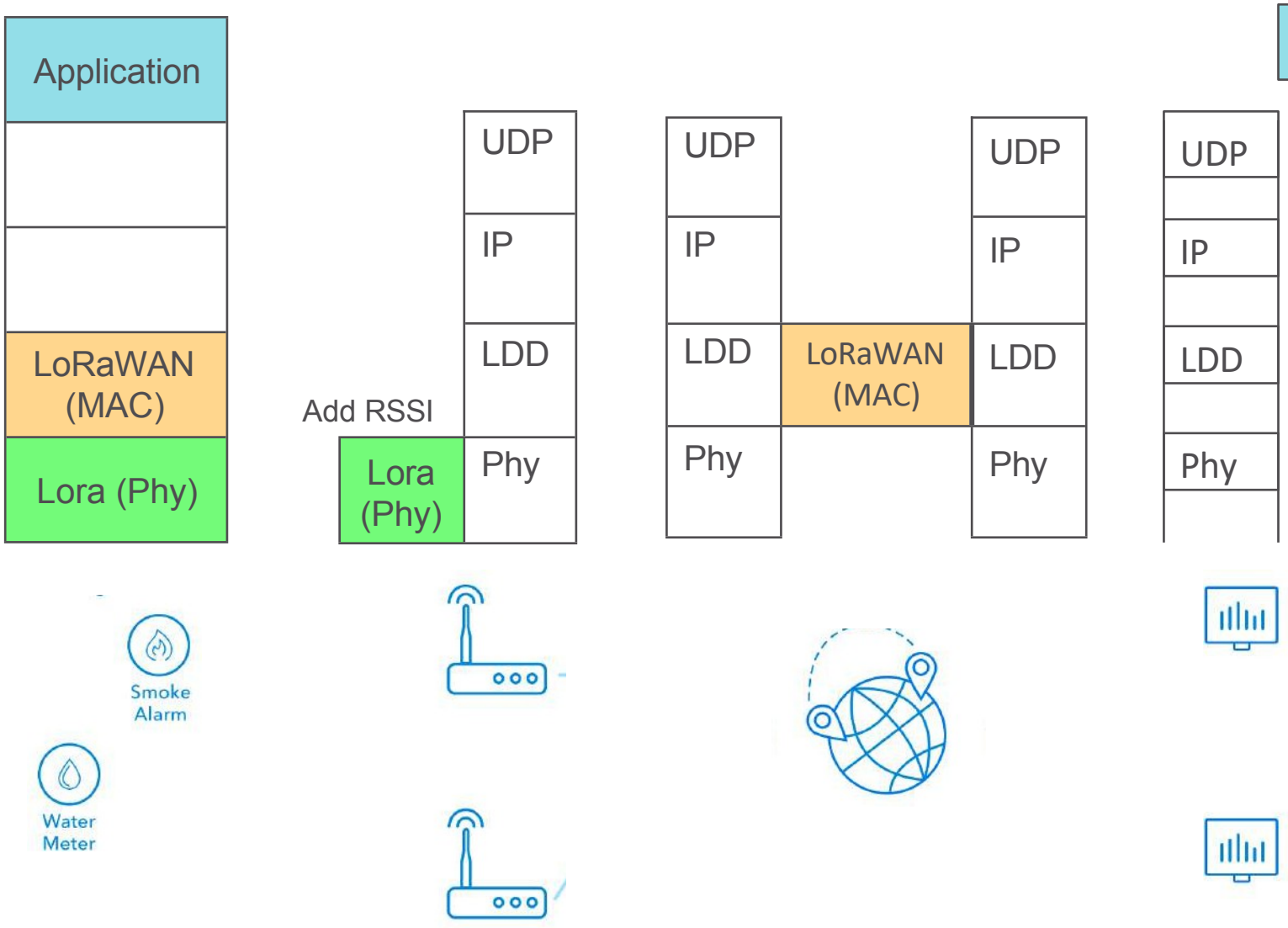
# LoRaWAN: Coverage



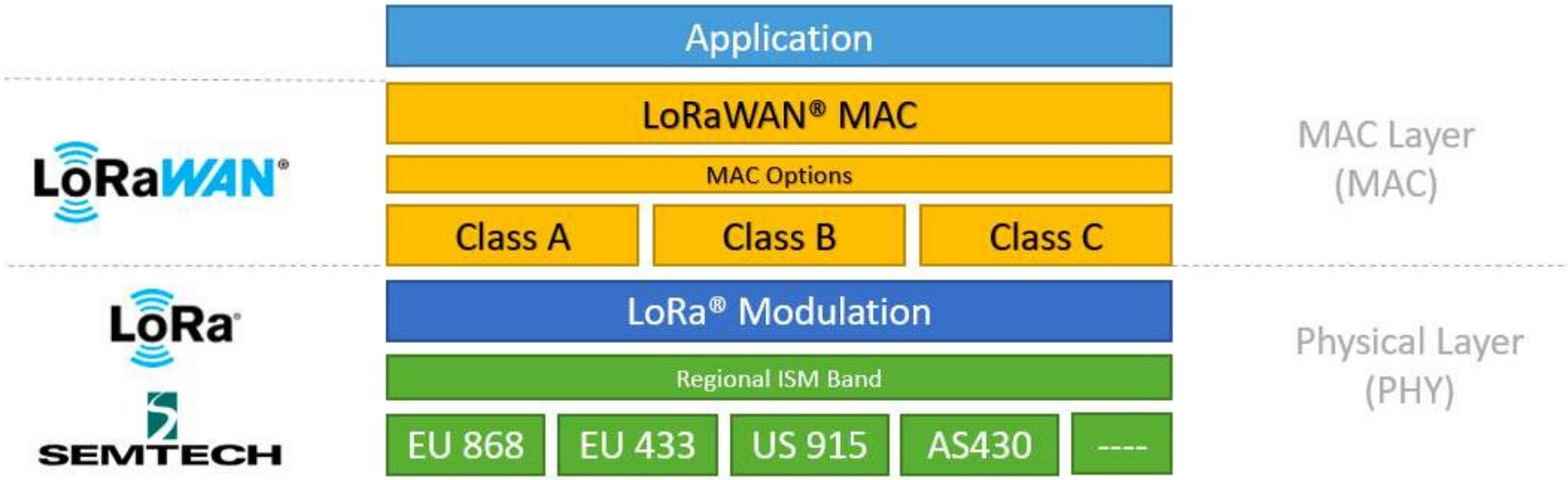
# LoRaWAN: Coverage



# LoRaWAN: Architecture and Protocol



# LoRa/LoRaWAN




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN: Data Rate

- 8 Standardized Data Rates

| DataRate | Configuration | Indicative physical bit rate [bit/s] |
|---|---|---|
| 0 | LoRa: SF12 / 125 kHz | 250 |
| 1 | LoRa: SF11 / 125 kHz | 440 |
| 2 | LoRa: SF10 / 125 kHz | 980 |
| 3 | LoRa: SF9 / 125 kHz | 1760 |
| 4 | LoRa: SF8 / 125 kHz | 3125 |
| 5 | LoRa: SF7 / 125 kHz | 5470 |
| 6 | LoRa: SF7 / 250 kHz | 11000 |
| 7 | FSK: 50 kbps | 50000 |
| 8..15 | RFU | |

Source: **LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018**

# LoRaWAN: Frequency Plan

## EU863-870

Uplink:

1. **868.1** - SF7BW125 to SF12BW125
2. **868.3** - SF7BW125 to SF12BW125 and SF7BW250
3. **868.5** - SF7BW125 to SF12BW125
4. **867.1** - SF7BW125 to SF12BW125
5. **867.3** - SF7BW125 to SF12BW125
6. **867.5** - SF7BW125 to SF12BW125
7. **867.7** - SF7BW125 to SF12BW125
8. **867.9** - SF7BW125 to SF12BW125
9. **868.8** - FSK

Downlink:

- Uplink channels 1-9 (RX1)
- **869.525** - SF9BW125 (RX2)

Source: TTN according to LoRaWAN regional parameters document

# LoRaWAN: Protocol – Channel access

- No device-to-device communications!
- →Only device-to→ -to-gateway and gateway-to-device communication
- No attachment to a gateway or listening
- Access based on a random mechanism (**Aloha**)
- Reminder: Two signals sent at the same time with two different SFs can be decoded without collision
- A logical channel = physical channel + spreading factor

# LoRaWAN: ADR

- **ADR: Adaptive Data Rate**
- At the request of nodes (devices)
- Allows the DR of the nodes to be adjusted to provide the highest possible throughput to the node
- The network server calculates this optimal throughput based on several measurements (last 20 measurements on TTN)
- Fastest throughput --> less transmission time →

  energy savings
- Caution: only recommended when radio conditions are very stable (often fixed nodes), otherwise this mode is disabled

# LoRaWAN: What about the downlink?

- Three node classes: Class A

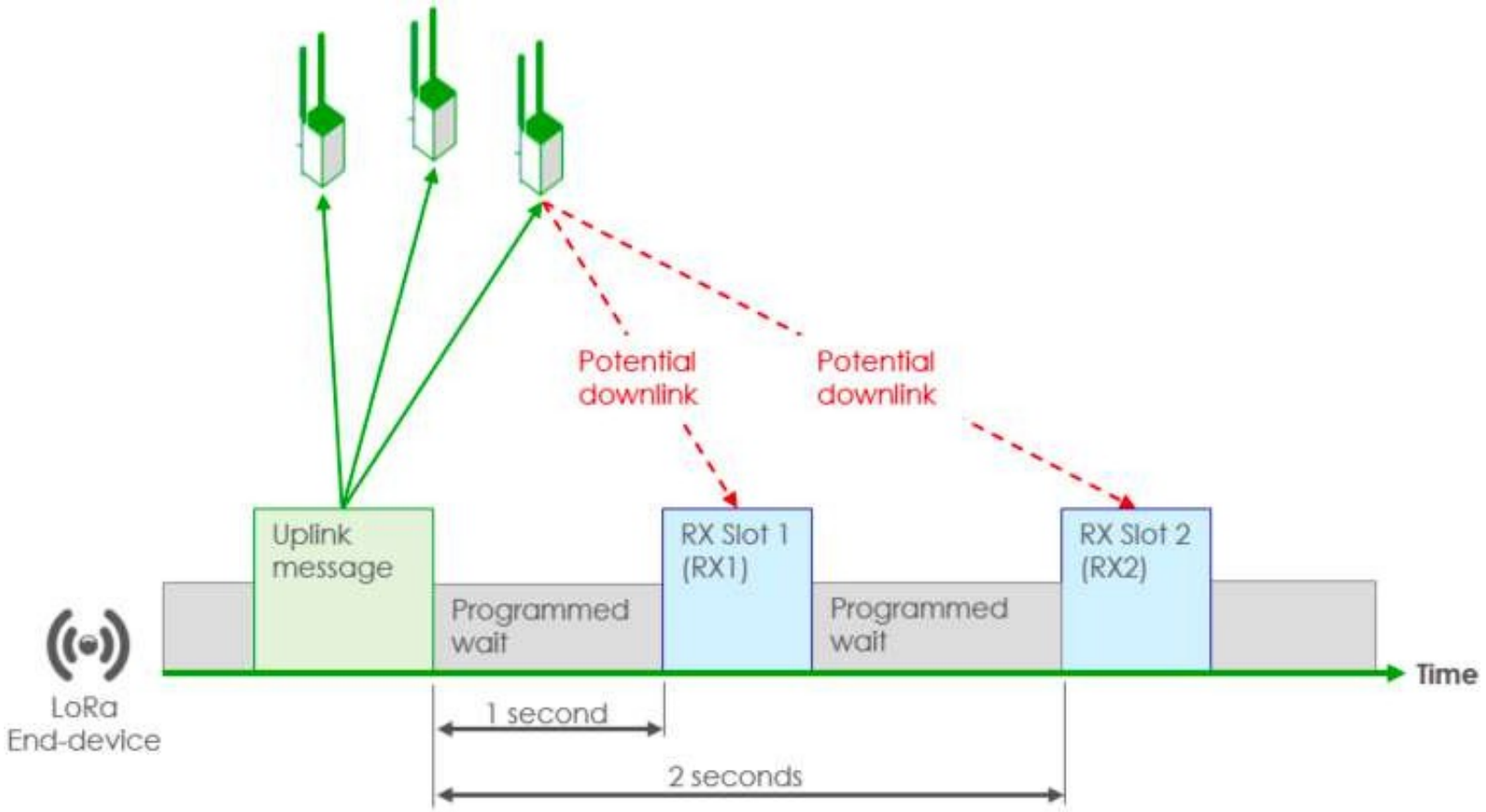



Image source: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

# LoRaWAN: What about the downlink?

- Three node classes: Class B

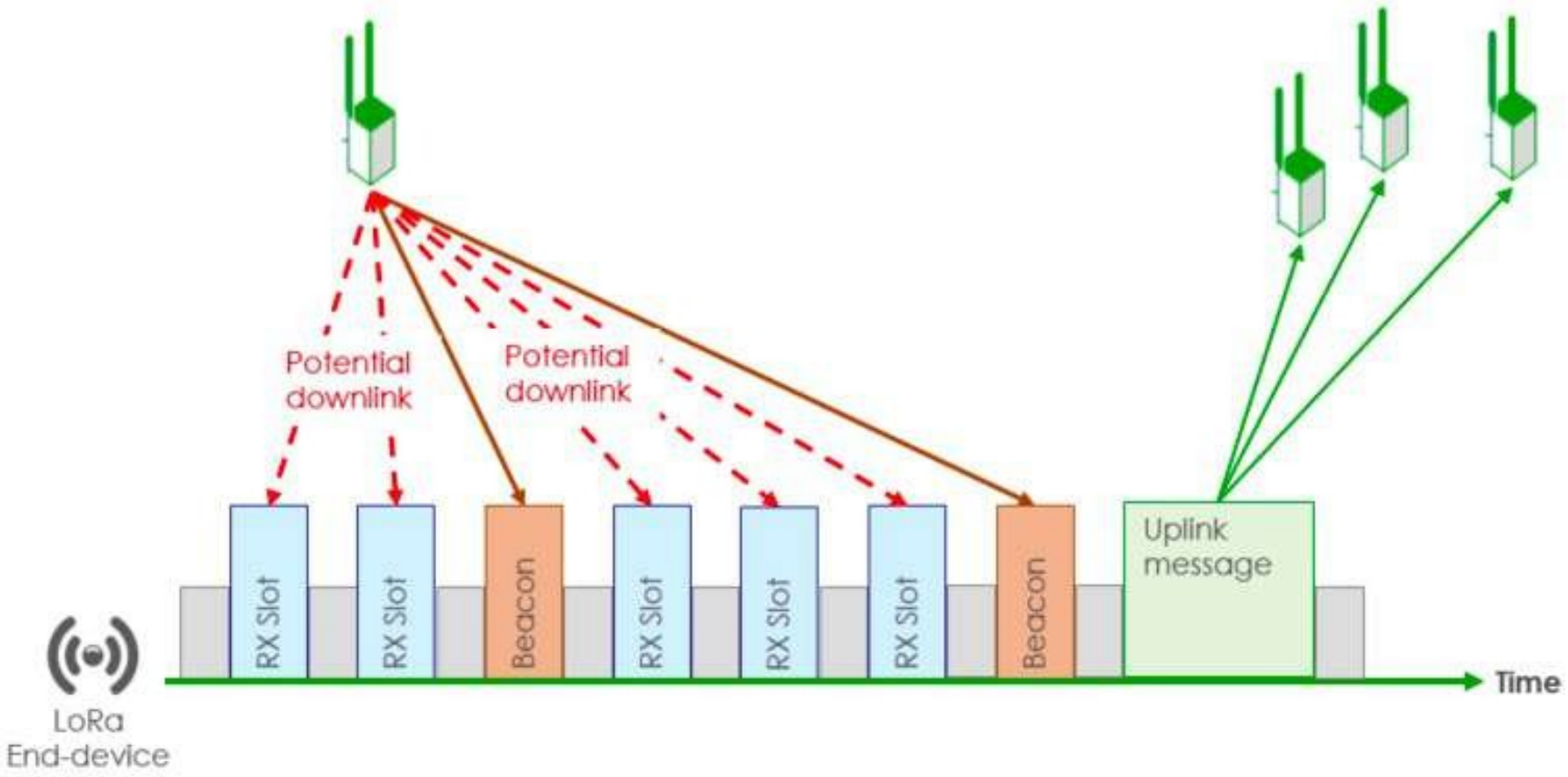



image source: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

# LoRaWAN: What about the downlink?

- Three node classes: Class C

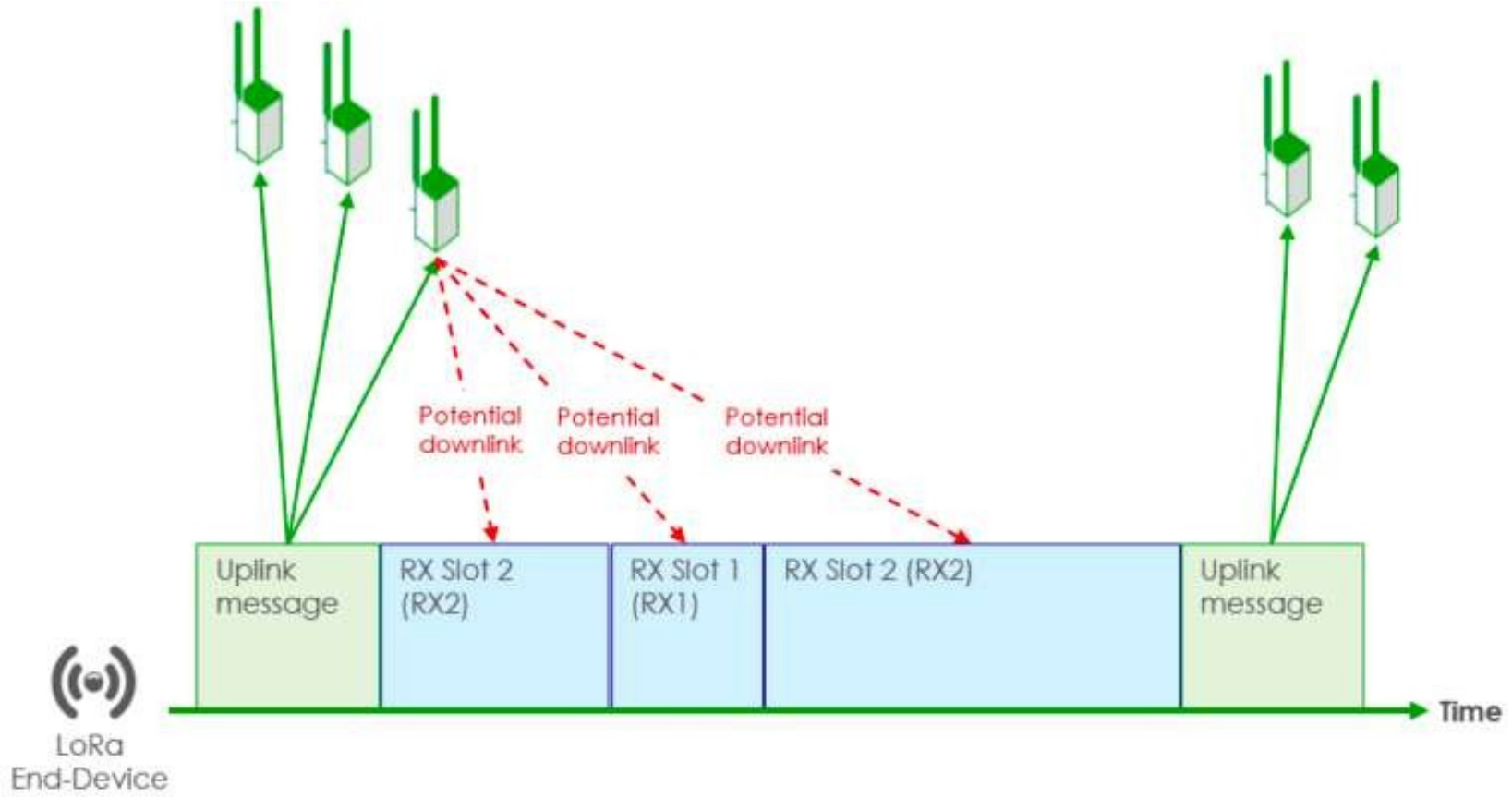
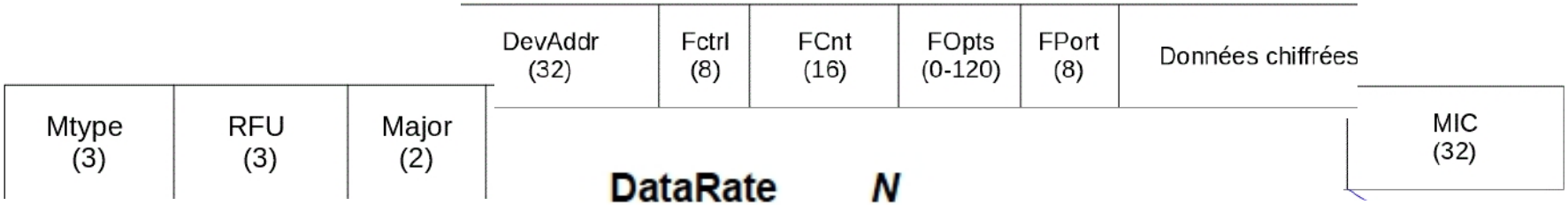



image source: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

# LoRaWAN: Packet format and maximum size




| DataRate | *N* |
|---|---|
| 0 | 51 |
| 1 | 51 |
| 2 | 51 |
| 3 | 115 |
| 4 | 222 |
| 5 | 222 |
| 6 | 222 |
| 7 | 222 |

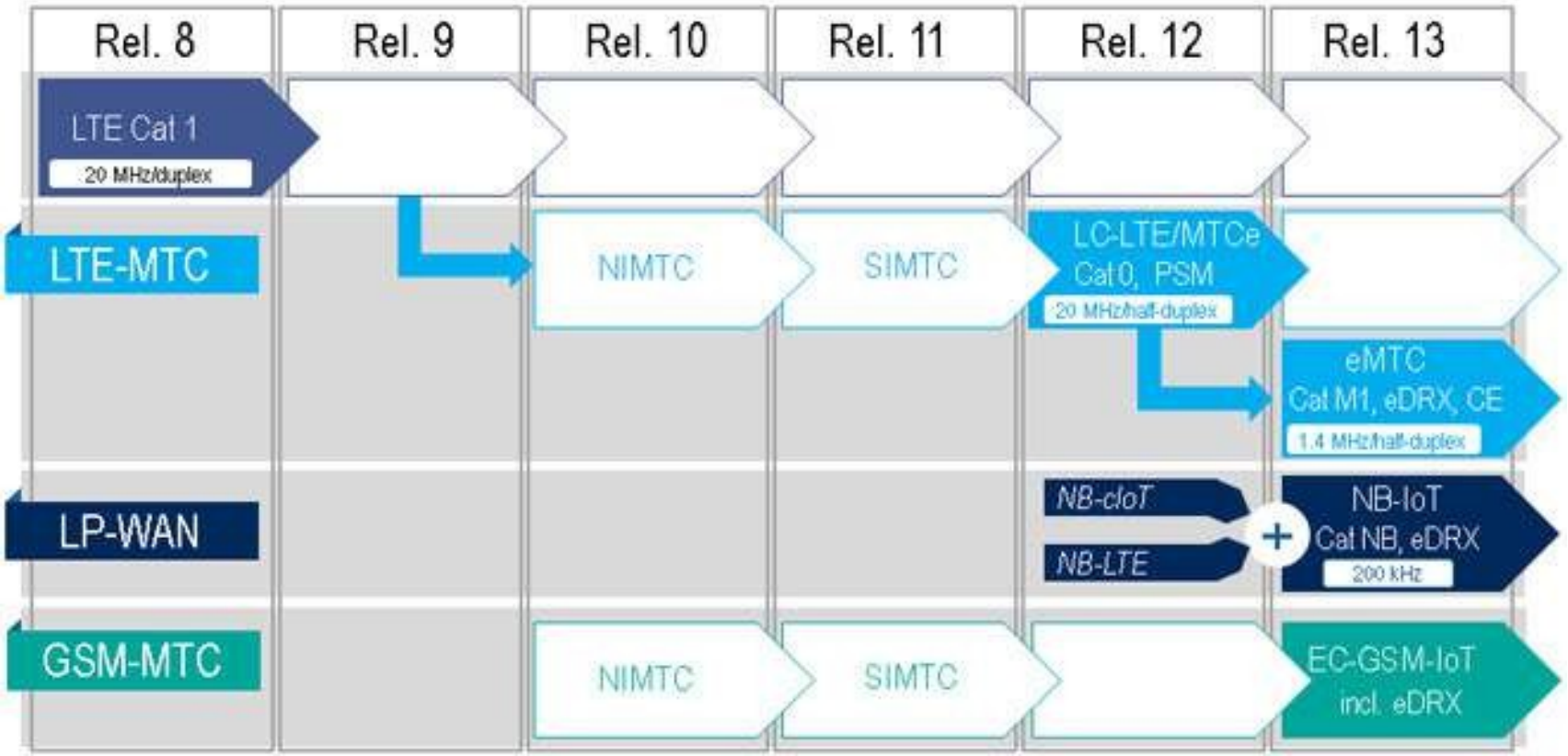
Source: **LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018**

⚠ Please note: in addition to the constraints related to duty cycles and message sizes, the operator may impose stricter constraints.

**Example TTN limits:**

Uplink: 30 seconds/day/node  Downlink: 10 messages/day/node

# Licensed frequencies: 3GPP IoT




**NB-IOT: Main actors:** Intel, Nokia, Ericsson, Huawei, Gemalto, etc.

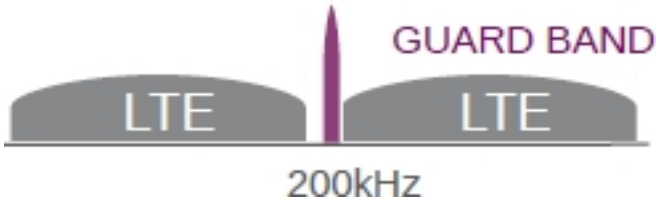
# Licensed frequencies: 3GPP IoT

| Standard (release 13) | eMTC(LTE-M) | NB-IOT | EC-GSM-LTE |
|---|---|---|---|
| Deployment | InBand LTE | **In-band LTE, guard-band LTE, standalone** | In-band GSM |
| Rate | 1Mbps UL/DL | **250 kbps DL 20/250 kbps UL** | 4 slots 70 kbps (GMSK) 240 (8PSK) |
| Bandwidth | 1.08 MHz | **200 kHz** | 200 kHz |



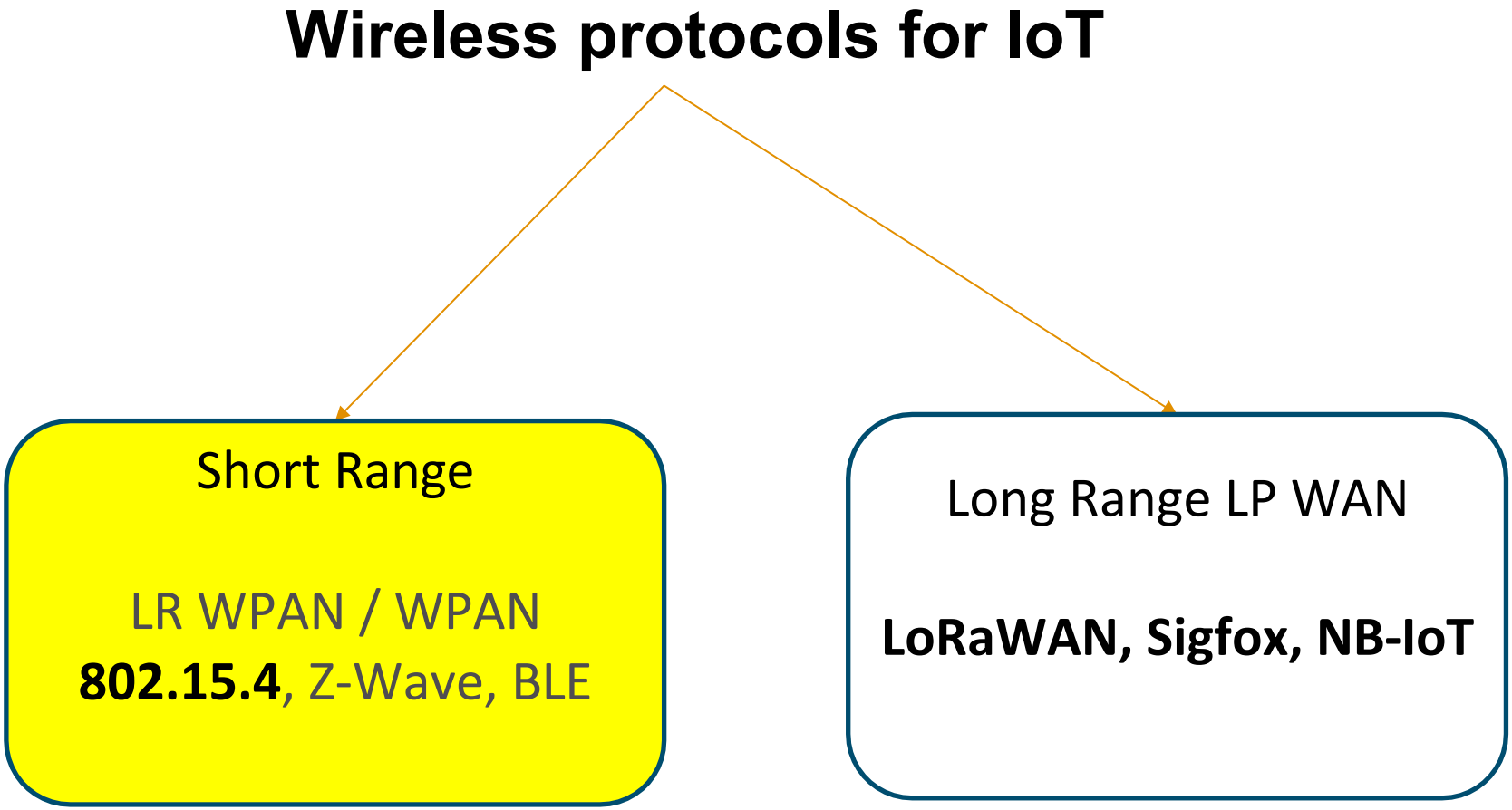




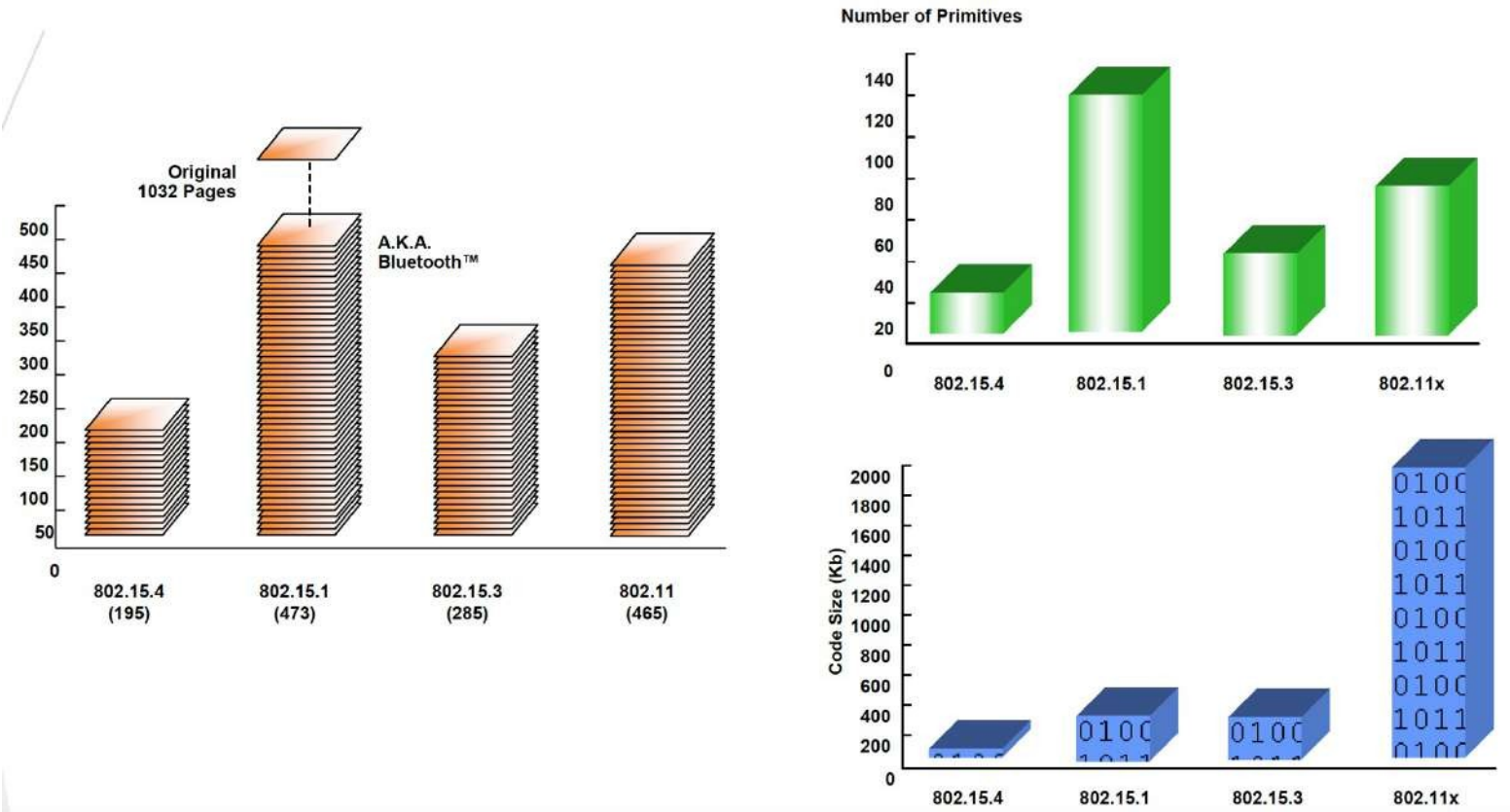


# 1.4 Short-range solutions

# Classification of IoT protocols



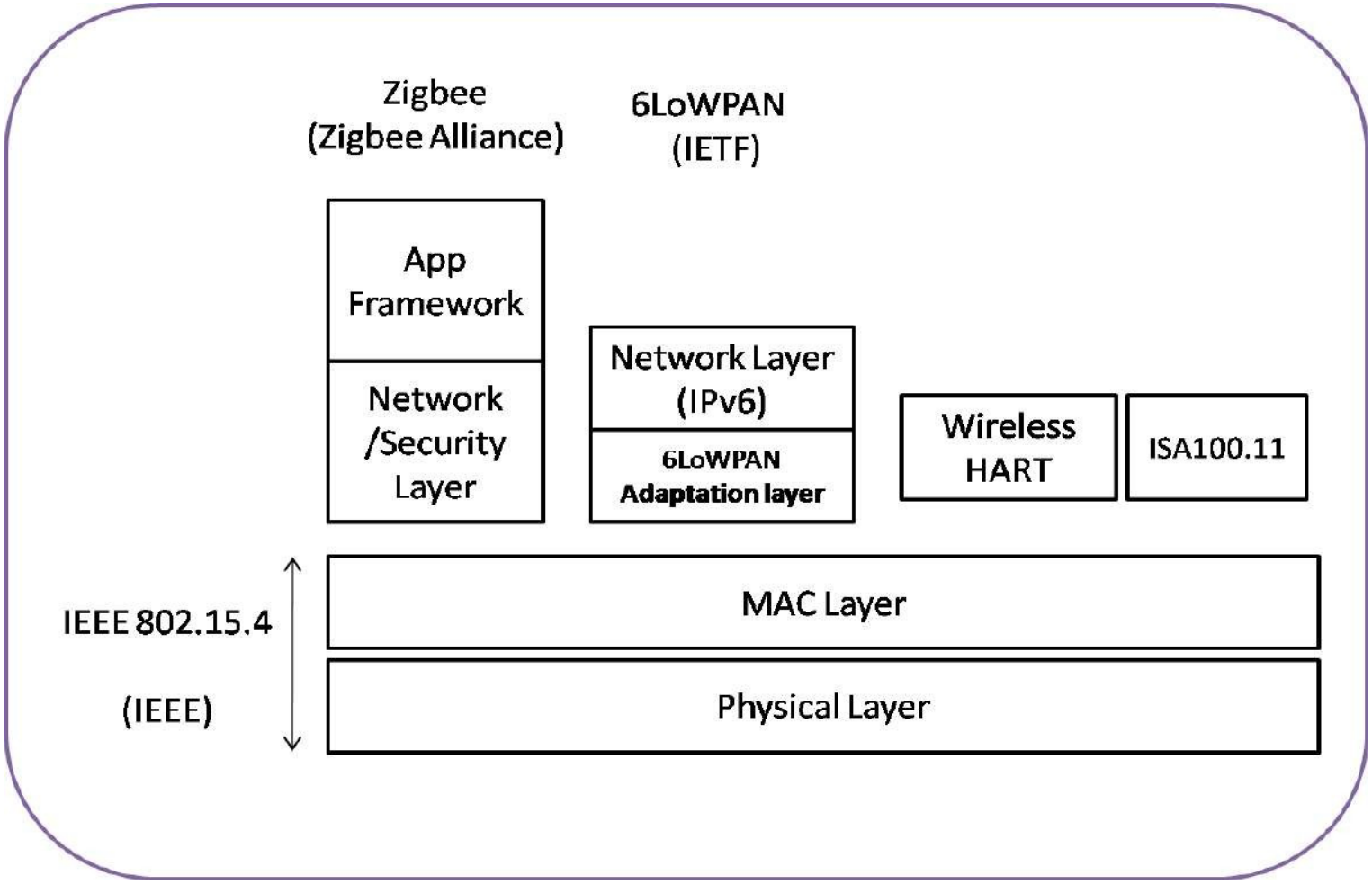
# Classification of the main standards




Source of figure: José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015

# LR WPAN: IEEE 802.15.4 and its positioning

## Example: IEEE 802.15.4



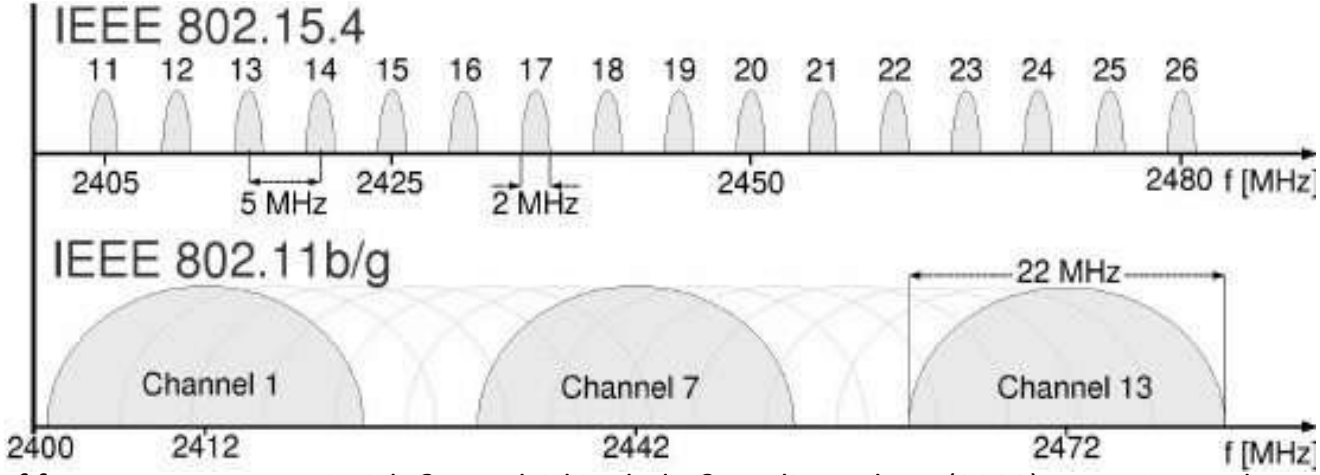
# IEEE 802.15.4 Physical Layer

- Low-complexity, low-power, low-cost **physical layer** (but low throughput ☺)
- **Unlicensed bands:**

  868-868.6 MHz in Europe (1 channel, 20 kbps)

  902-928 MHz in America (10 channels, 40 kbps)

  2400-2483.5 MHz worldwide (16 channels, 250 kbps)



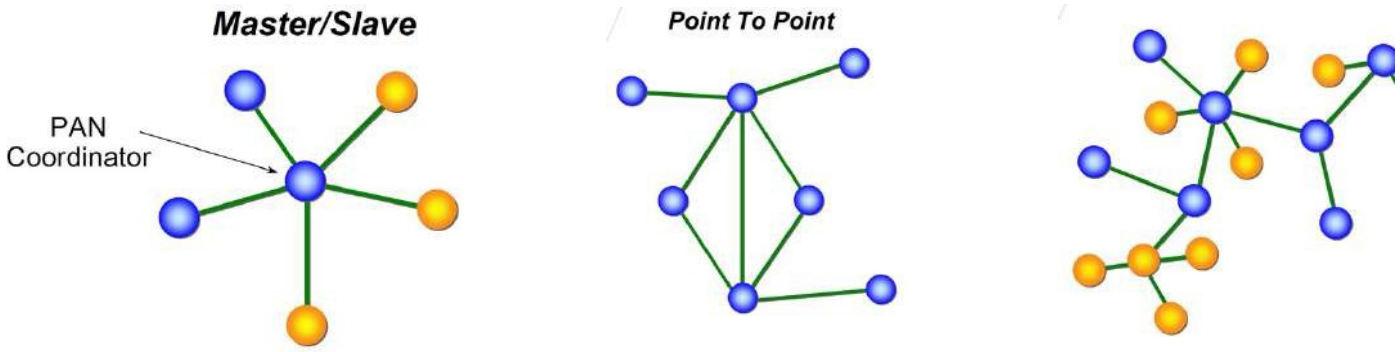

Source of figure: Hauer, Jan-Hinrich & Handziski, Vlado & Wolisz, Adam. (2009). Experimental Study of the Impact of WLAN Interference on IEEE 802.15.4 Body Area Networks. 17-32.

# 802.15.4 MAC layer (types and topologies)

- Two classes of devices:

  **FFD (full function device)**: can coordinate and communicate with any other node

  **RFD (Reduced Function Device):** simple implementation, can only communicate with an FFD.




Three types of nodes: PAN coordinator (FFD), coordinator (FFD), and network device (RFD or FFD)

Source of figure: José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015

# 802.15.4 MAC layer (identifiers and frame)

- **EUI-64 addresses:** Each device that supports 802.15.4 must have a globally unique EUI-64 address (equivalent to a MAC address).
- Devices are grouped into PANs identified by two bytes
- Each device can be configured with a short 2-byte address

# 802.15.4 MAC Layer (Identifiers and Frame)

| Bits: 0–2 | 3 | 4 | 5 | 6 | 7–9 | 10–11 | 12–13 | 14–15 |
|---|---|---|---|---|---|---|---|---|
| Frame Type | Security Enabled | Frame Pending | AR | PAN ID Compression | Reserved | Dest. Addressing Mode | Frame Version | Source Addressing Mode |

| Octets: 2 | 1 | 0/2 | 0/2/8 | 0/2 | 0/2/8 | 0/5/6/10/14 | variable | 2 |
|---|---|---|---|---|---|---|---|---|
| Frame Control | Sequence Number | Destination PAN Identifier | Destination Address | Source PAN Identifier | Source Address | Auxiliary Security Header | Frame Payload | FCS |
| | | Addressing fields | | | | | | |
| MHR | | | | | | | MAC Payload | MFR |

127 bytes max

Source of figure: Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details: https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

# 802.15.4 MAC layer (identifiers and frame)

| Bits: 0–2 | 3 | 4 | 5 | 6 | 7–9 | 10–11 | 12–13 | 14–15 |
|---|---|---|---|---|---|---|---|---|
| Frame Type | Security Enabled | Frame Pending | AR | PAN ID Compression | Reserved | Dest. Addressing Mode | Frame Version | Source Addressing Mode |

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|---|---|---|---|---|---|---|---|---|
| Frame Control | Sequence Number | Destination PAN Identifier | Destination Address | Source PAN Identifier | Source Address | Auxiliary Security Header | Frame Payload | FCS |
| | | Addressing fields | | | | | | |
| MHR | | | | | | | MAC Payload | MFR |

Addressing fields

127 bytes max

Source of figure: Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details: https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

# 802.15.4 MAC layer (identifiers and frame)

| Bits: 0–2 | 3 | 4 | 5 | 6 | 7–9 | 10–11 | 12–13 | 14–15 |
|---|---|---|---|---|---|---|---|---|
| Frame Type | Security Enabled | Frame Pending | AR | PAN ID Compression | Reserved | Dest. Addressing Mode | Frame Version | Source Addressing Mode |

| Octets: 2 | 1 | 0/2 | 0/2/8 | 0/2 | 0/2/8 | 0/5/6/10/14 | variable | 2 |
|---|---|---|---|---|---|---|---|---|
| Frame Control | Sequence Number | Destination PAN Identifier | Destination Address | Source PAN Identifier | Source Address | Auxiliary Security Header | Frame Payload | FCS |
| | | Addressing fields | | | | | | |
| MHR | | | | | | | MAC Payload | MFR |

Security fields

127 bytes max

Source of figure: Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details: https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

# 802.15.4 MAC layer: Channel access

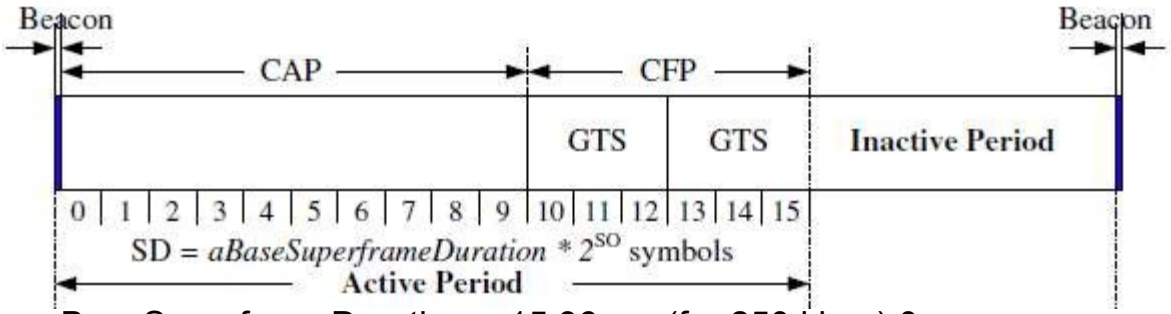
- **Non-beacon-enabled mode:** Non-slotted CSMA-CA
- **Beacon-enabled mode:** CSMA-CA "slotted"

  A superframe sent by the PAN coordinator:

  – A Contention Access Period (CAP): Slotted CSMA/CA;

  – Contention-Free Period (CFP): Guaranteed access slots (GTS)

  – An optional idle period

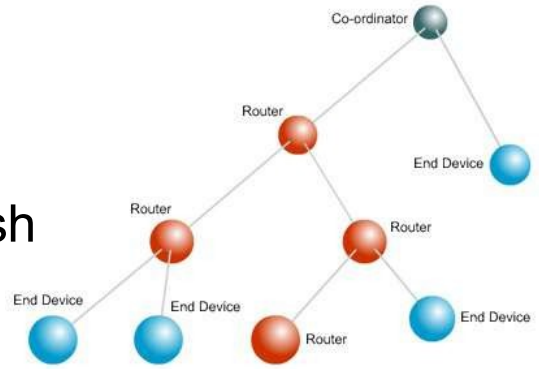



aBaseSuperframeDuration = 15.36 ms (for 250 kbps) 0 ≤ SFO ≤14

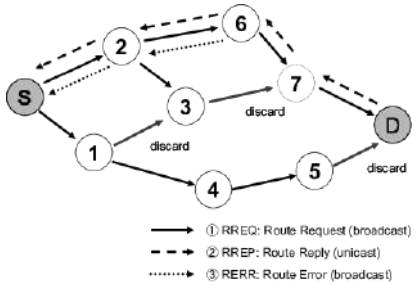
# 1.3 IEEE 802.15.4 short range solutions and upper layers

# Zigbee

- **Three types of nodes**
  - –- ZigBee coordinator (ZC): Initialization, maintenance, and control of the network
  - – ZigBee router (ZR): packet routing
  - – ZigBee end device (ZED): End sensors
- **Three types of topologies:** star, tree, and mesh



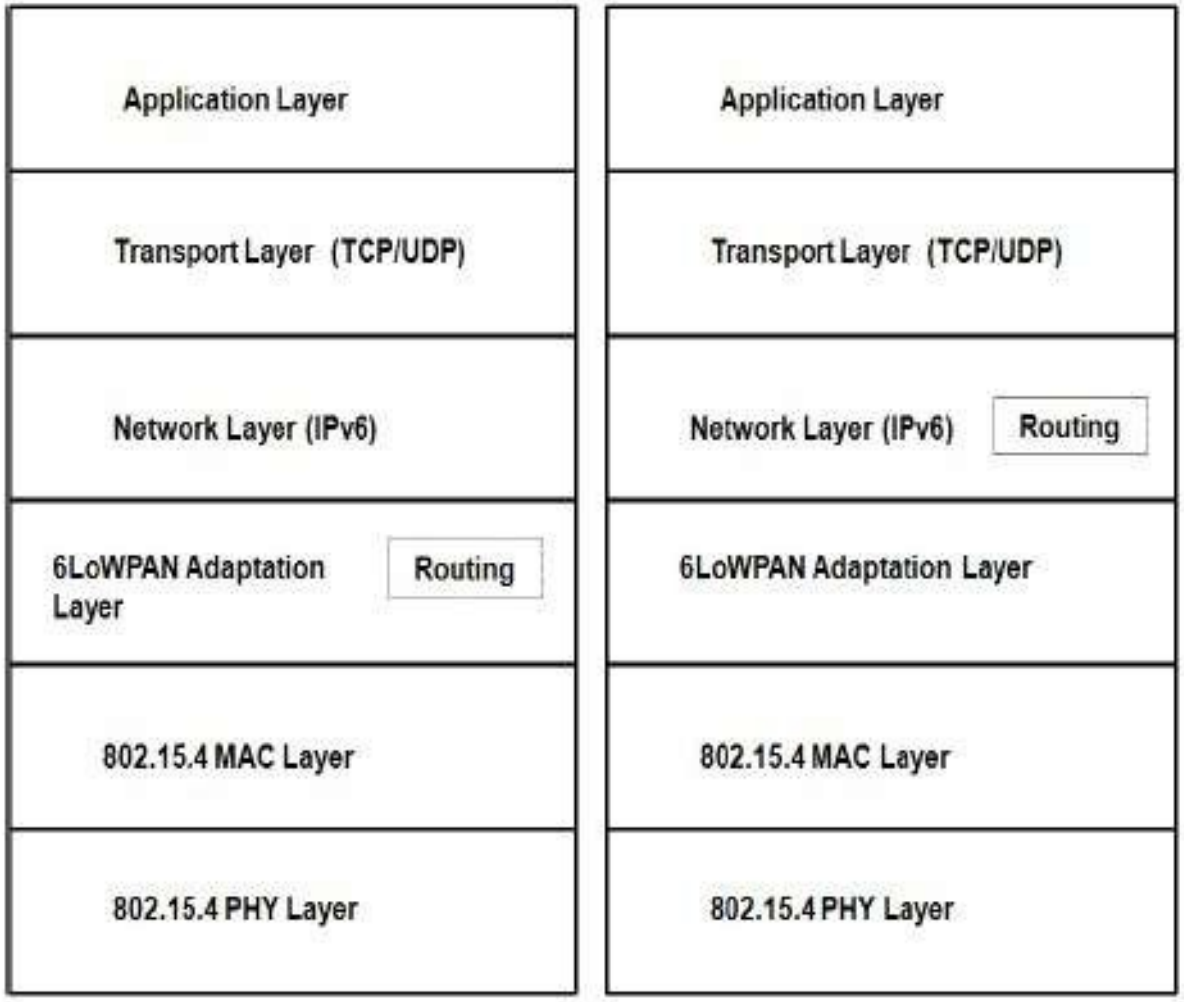

- **Tree topology:** routing based on assigned addresses so that the parent can be deduced automatically
- **Mesh topology:** routing through a reactive AODV protocol



# 6LowPAN

- Adaptation of IPv6 to support low-power, low-cost devices
- To be implemented on top of the IEEE 802.15.4 standard
- Adaptation layer for fragmenting IP packets into IEEE 802.15.4 frames (maximum 127 bytes)
- 6LowPan defines IPv6 header compressions
- 6LowPAN: Interoperability vs. complexity and delay

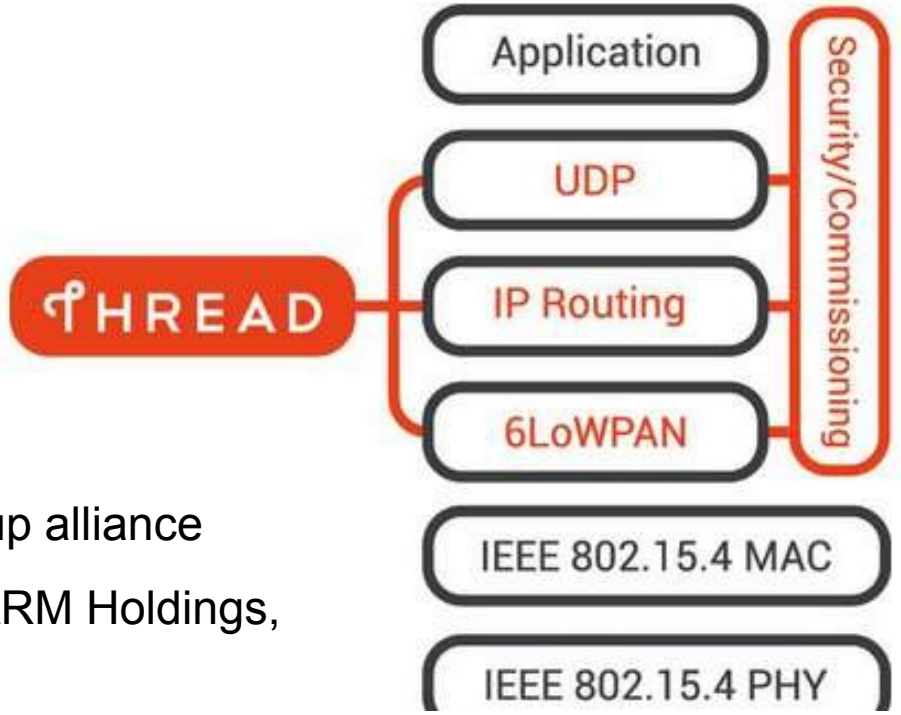
# 6LowPAN::Routing




6LoWPAN Mesh-under Routing

6LoWPAN Route-over Routing

# 6LowPAN & Thread

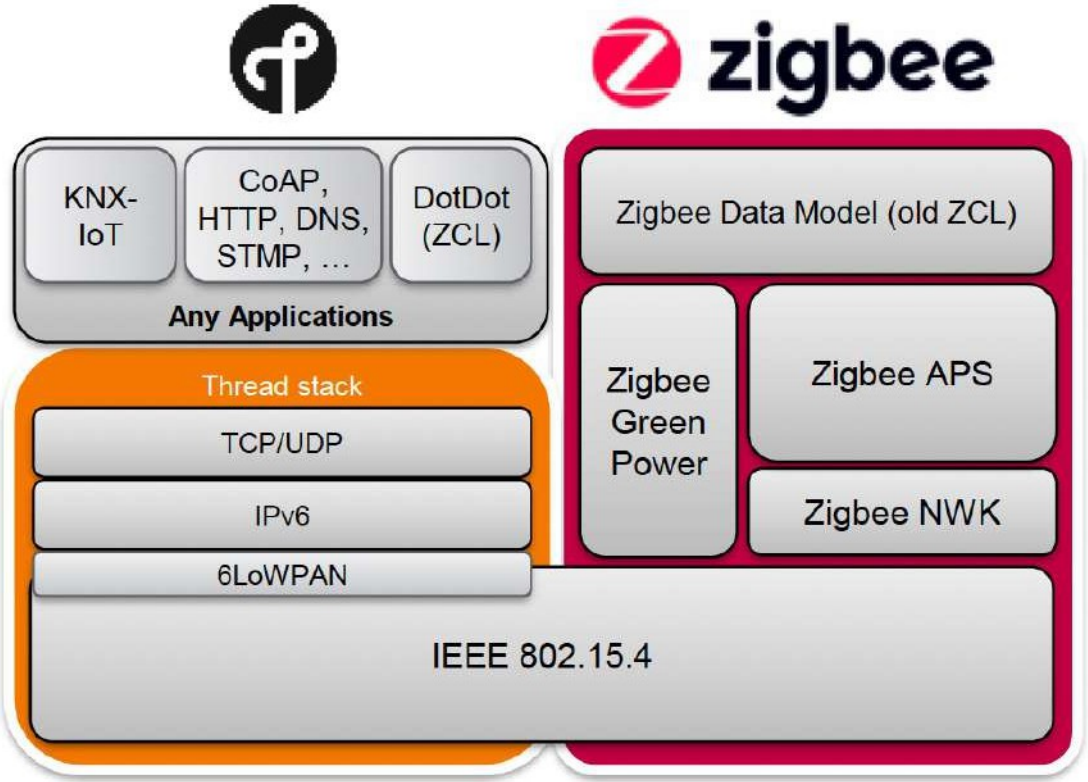



2014: Launch of the Thread Group alliance

Google (Nest Labs), Samsung, ARM Holdings, Qualcomm, etc.

2018: Apple joins the group...

# Thread vs. Zigbee




Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy, Nicolas Cochard, TC, 2021

# Matter project

- **December 2019**, Amazon, Apple, Google, Samsung SmartThings, the Zigbee Alliance→ Project Connected Home over IP (CHIP)
- **May 2021:** Zigbee Alliance→ Connectivity Standards Alliance

  Project Connected Home over IP (CHIP)→ **Matter**

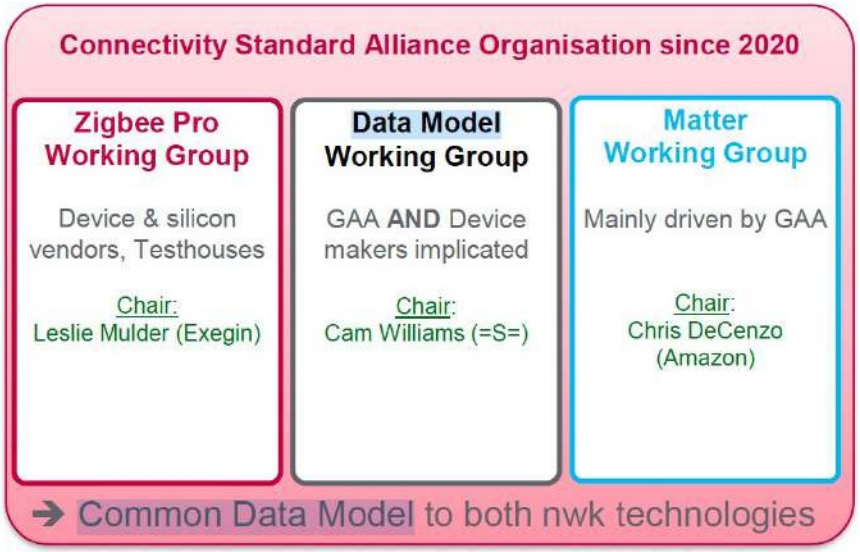
Matter: IP-based with an open-source approach 09/30/2022 -
Release of version 1.0 of the Matter standard 10/23/2023 -
Release of version 1.2 of the Matter standard




Source: https://buildwithmatter.com/ ☺

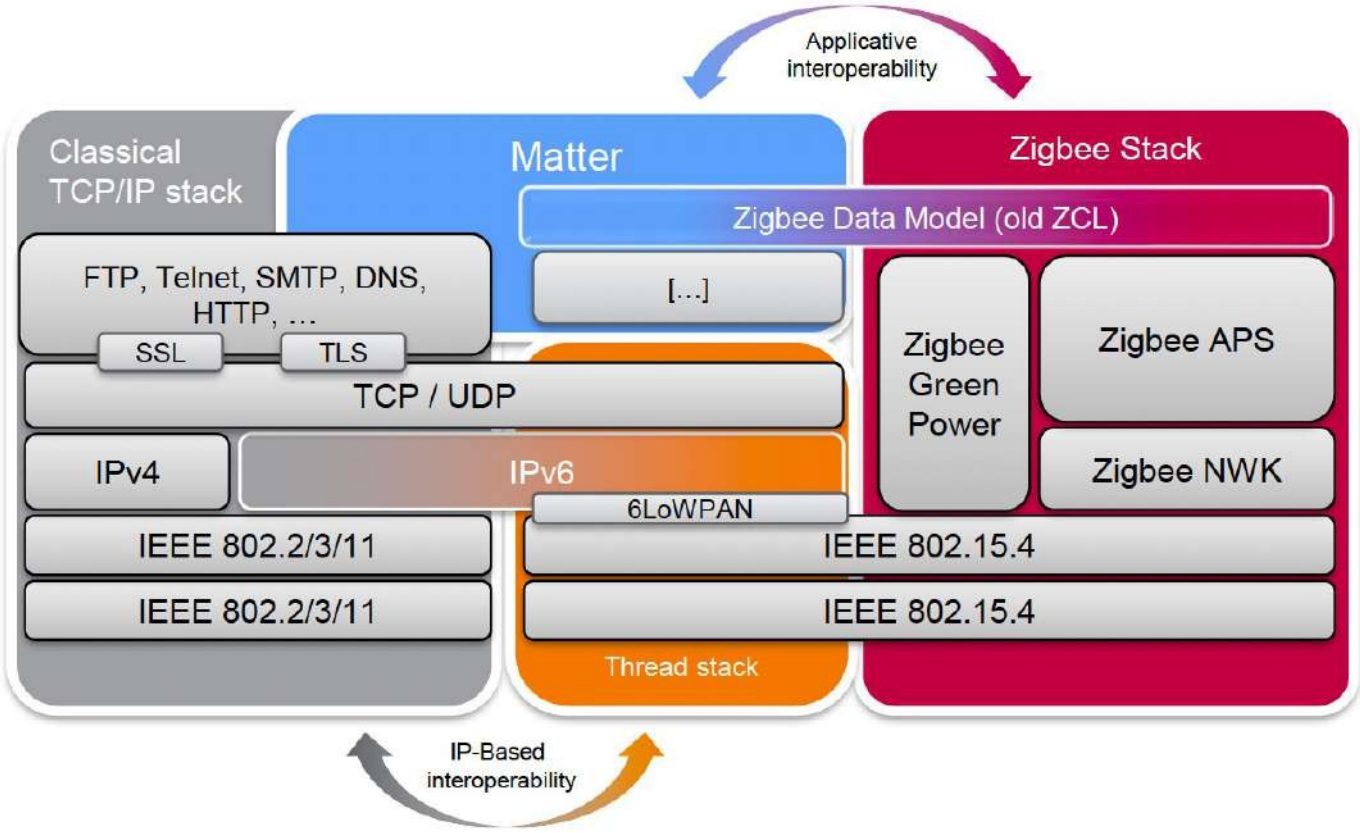
# CSA positioning




Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy, Nicolas Cochard, TC, 2021

# Mafler, Thread, and Zigbee




Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy, Nicolas Cochard, TC, 2021

# Main references


- José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015.
- Hauer, Jan-Hinrich & Handziski, Vlado & Wolisz, Adam. (2009). Experimental Study of the Impact of WLAN Interference on IEEE 802.15.4 Body Area Networks. 17-32.
- Dedicated networks for IoT: PHY/MAC state of the art and challenges C. Goursaud, JM. Gorce, EAI Endorsed Transactions on the Internet of Things, 2016
- Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.
- Training Course LoRa / LoRaWAN, Sylvain MONTAGNY, University of Savoie Mont Blanc
- Characterization of processes modulation modulation and modulation demodulation LoRa, J-F SERREAU, 2019
- http://genelaix.free.fr/IMG/pdf/presentation_lora_lorawan.pdf
- http://www.onditech.com/documents/freq_433_868.pdf

# Main references


- LoRaWAN Specification v1.1, LoRa Alliance, 2017
- LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018
- Training courses LoRa / LoRaWAN, Sylvain MONTAGNY, University of Savoie Mont Blanc
- Augustin, Aloÿs & Yi, Jiazi & Clausen, Thomas Heide & Townsley, William. (2016). A Study of LoRa: Long Range & Low Power Networks for the Internet of Things
- https://www.thethingsnetwork.org/
- http://genelaix.free.fr/IMG/pdf/presentation_lora_lorawan.pdf
- http://www.onditech.com/documents/freq_433_868.pdf
- https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972
- https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# 2- IoT security

# Security challenges in WSN

## Information Security

$Risk = Threat \times Vulnerability \times Cost$

### Vulnerability

- Deployment environment
- Lack of physical protection
- Memory limitation
- Energy limitation
- Wireless Communication
- Lack of infrastructure

### Threat

- Traditional threats (eavesdropping, message injection ...)
- New specific attacks (energy drain, data aggregation attacks)

### Cost

- Critical applications of WSN
- Dramatic effects and costs

Ex: Mirai botnet (DDoS attacks in 2016/17)

# Security challenges in WSN

**IT life cycle**

- Factorized software
- Upgrades
- Soft restes when unusable
- Large scale
- Mature codebase

**IoT life cycle**

- Fragmentation
- No upgrades
- Non-soft restes
- Very large scale
- Young codebase

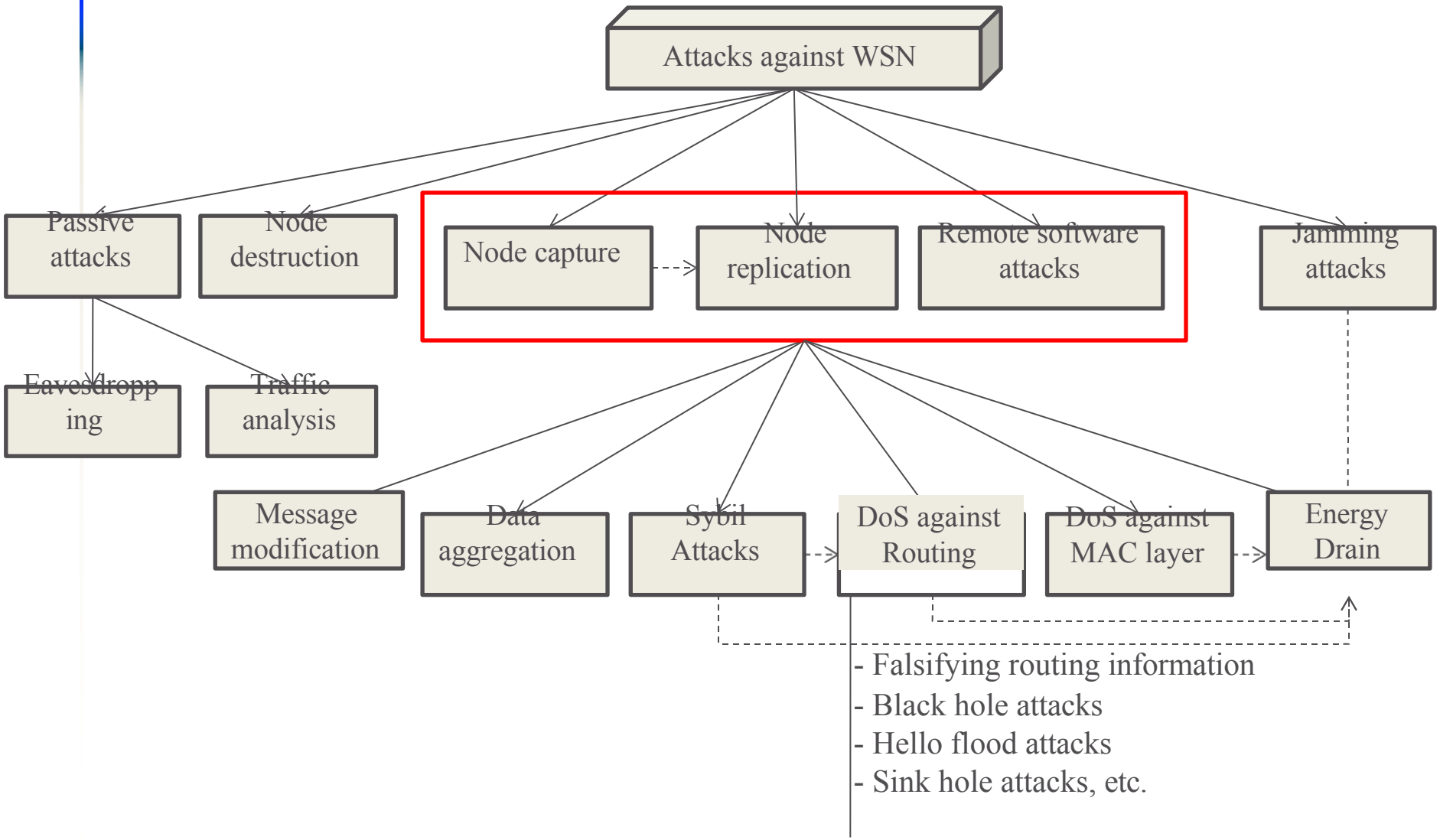
Source: How IoT revisits security problems, François Lesueur, INSA-Lyon, April 2017

It's not the walls that protect the citadel, but the spirit of its inhabitants."

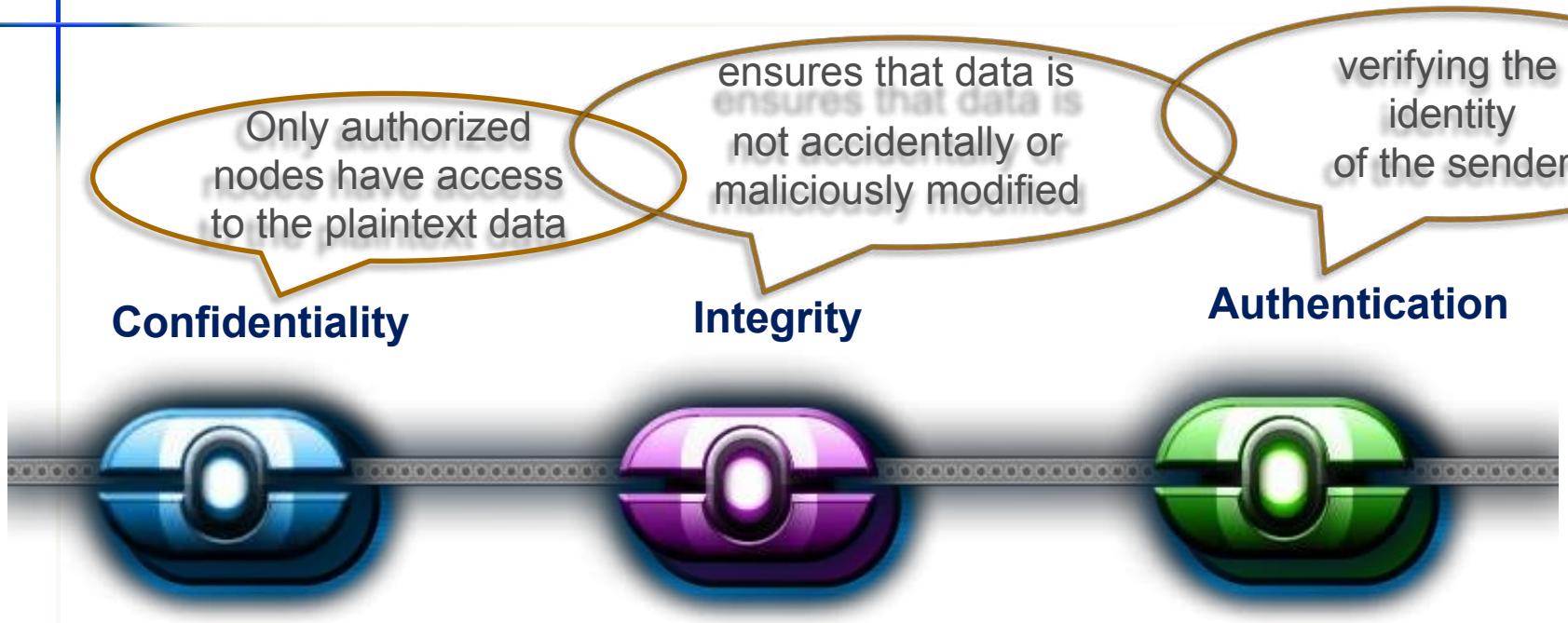
[Thucydides]

At least 40% of attacks are caused by users themselves.

# Attacks against WSN



# Security requirement in WSN




- **Basic services**
- **Securing routing**
- **Securing data aggregation**



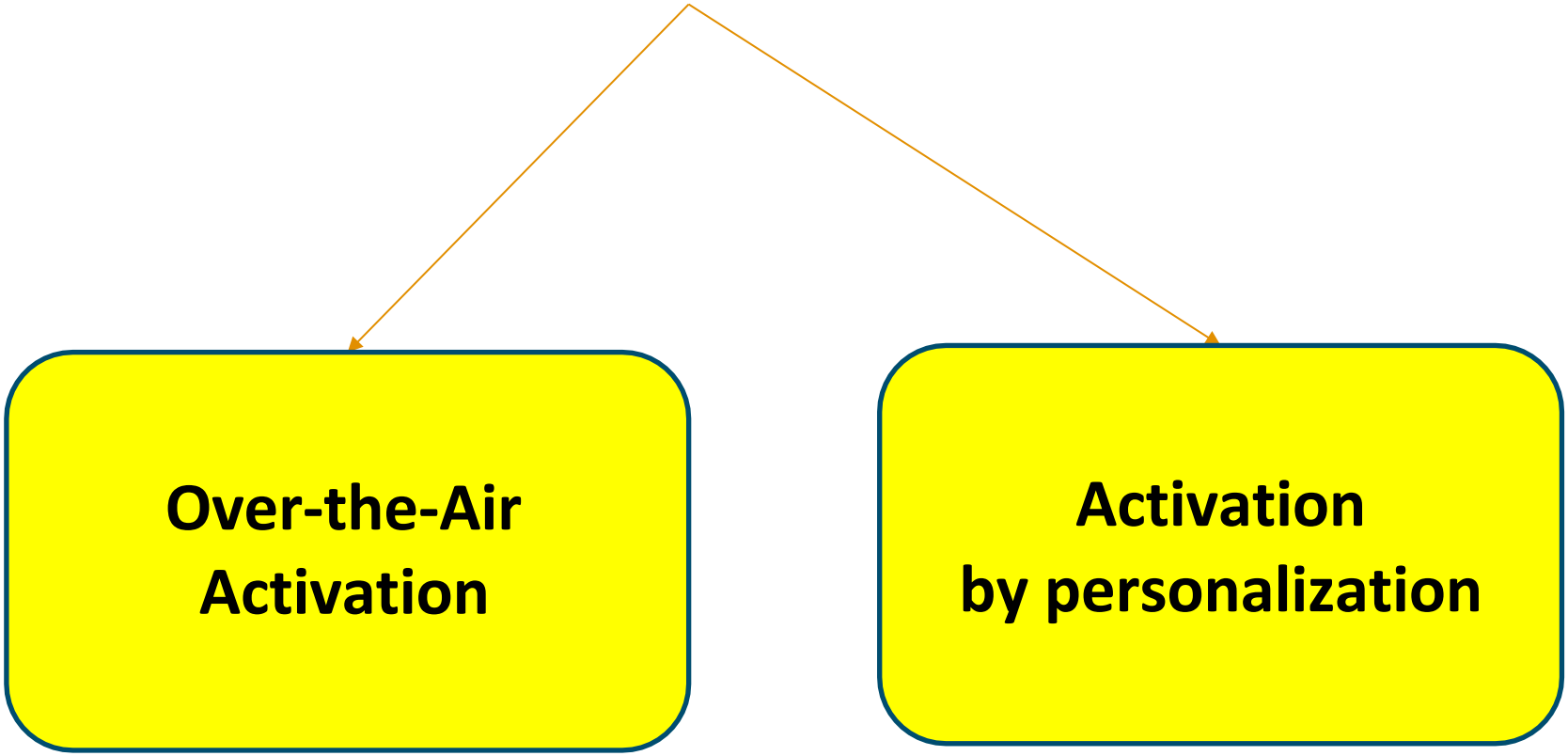
**Key management:**

**A cornerstone**

# 2- IoT security

# 2.1 LoRaWAN networks

# LoRaWAN security



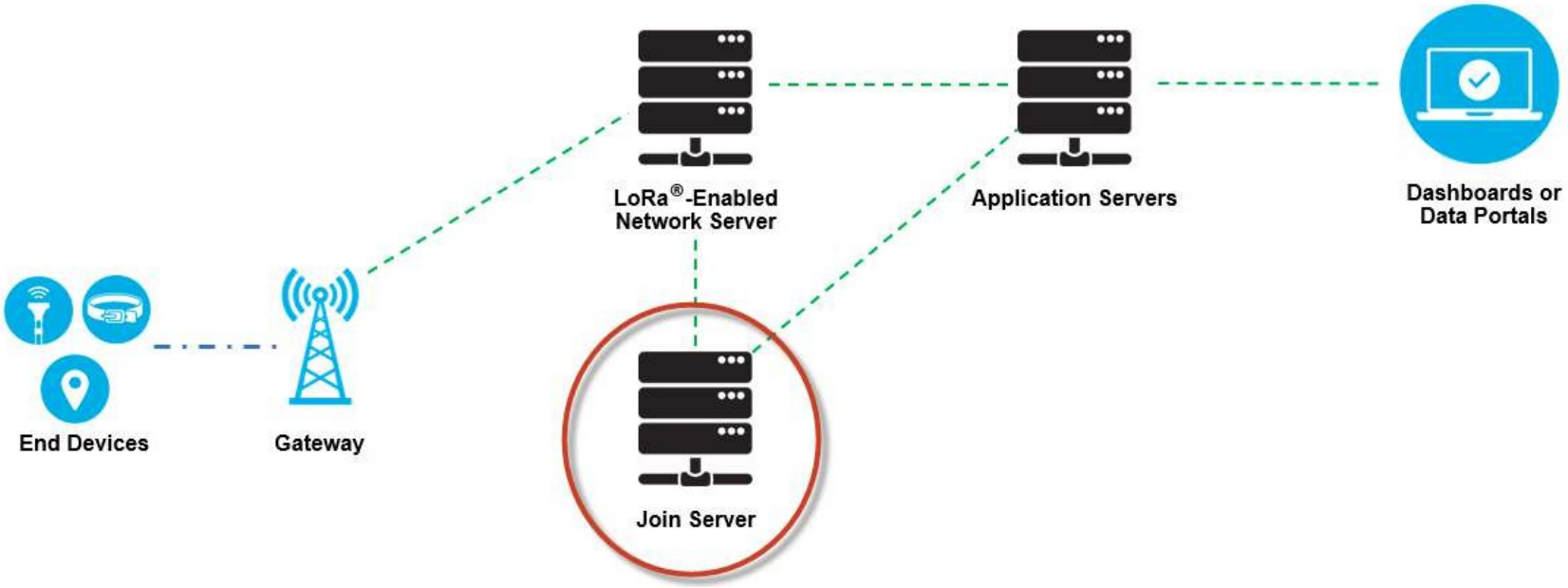
# LoRaWAN Security

## LoRaWAN security: Two modes

- **OTAA (Over-the-Air Activation)**
  - 128-bit Master Key (**Appkey**) for initial mutual authentication and the generation of two **session keys**:
    - Network Session Key (**NwkSKey**) for message integrity and authentication
    - Application Session Keys (**AppSKey**) for encryption (confidentiality)
- **ABP (Activation by personalization)**
  - Each end node is provided with **NwkSKey and AppSKey**

# LoRaWAN Security: OTTA

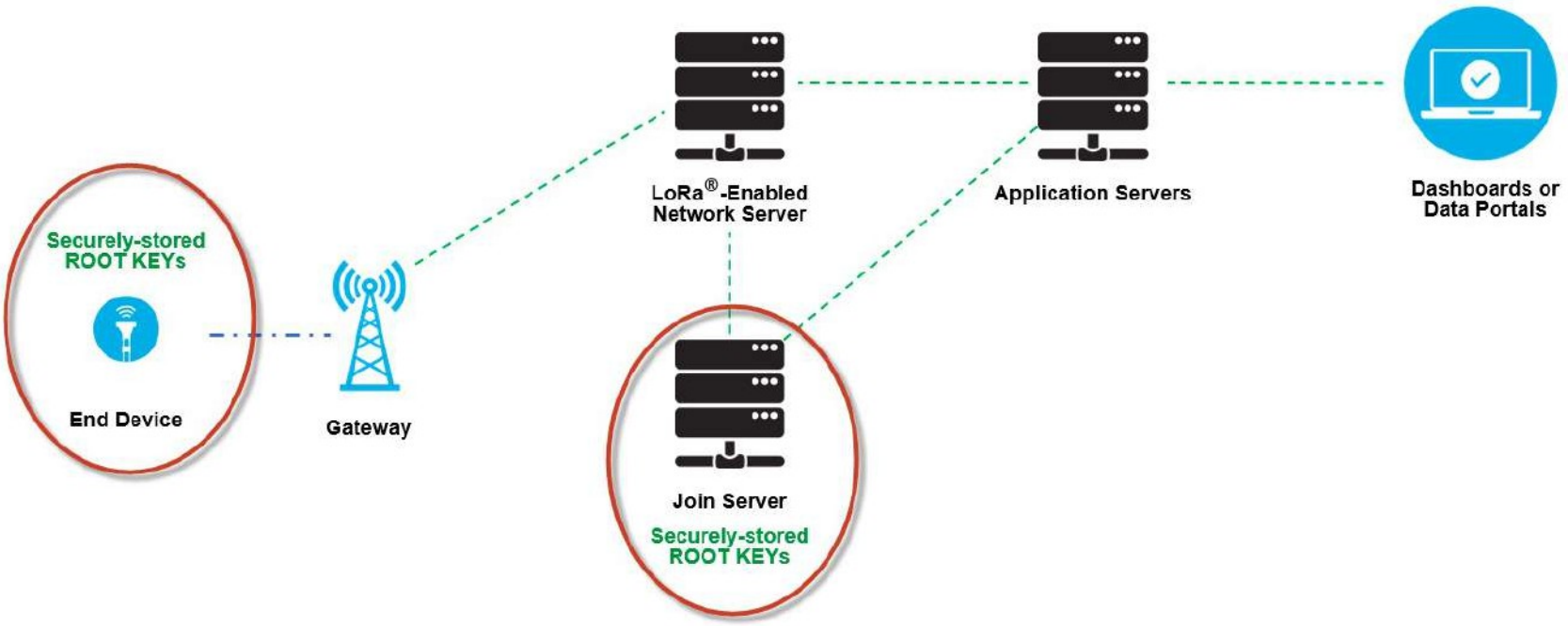
## Join Server:




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN Security: OTTA

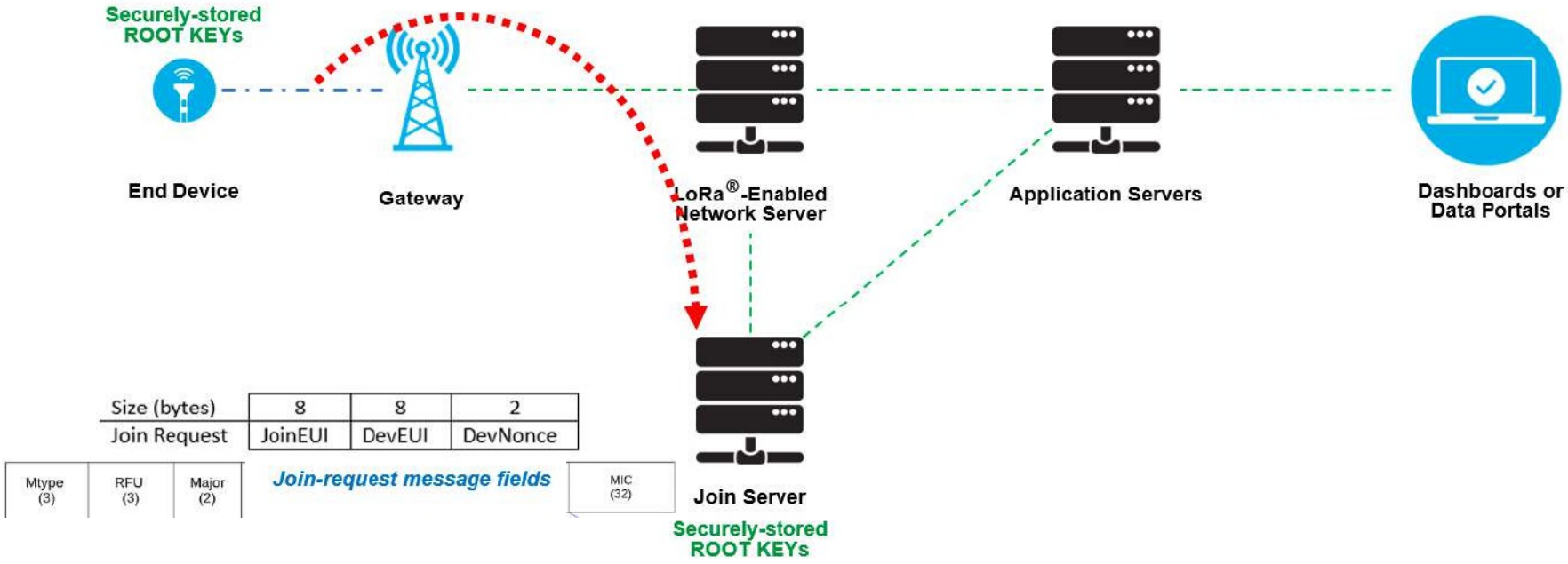
## Root key: AppKey




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN Security: OTTA

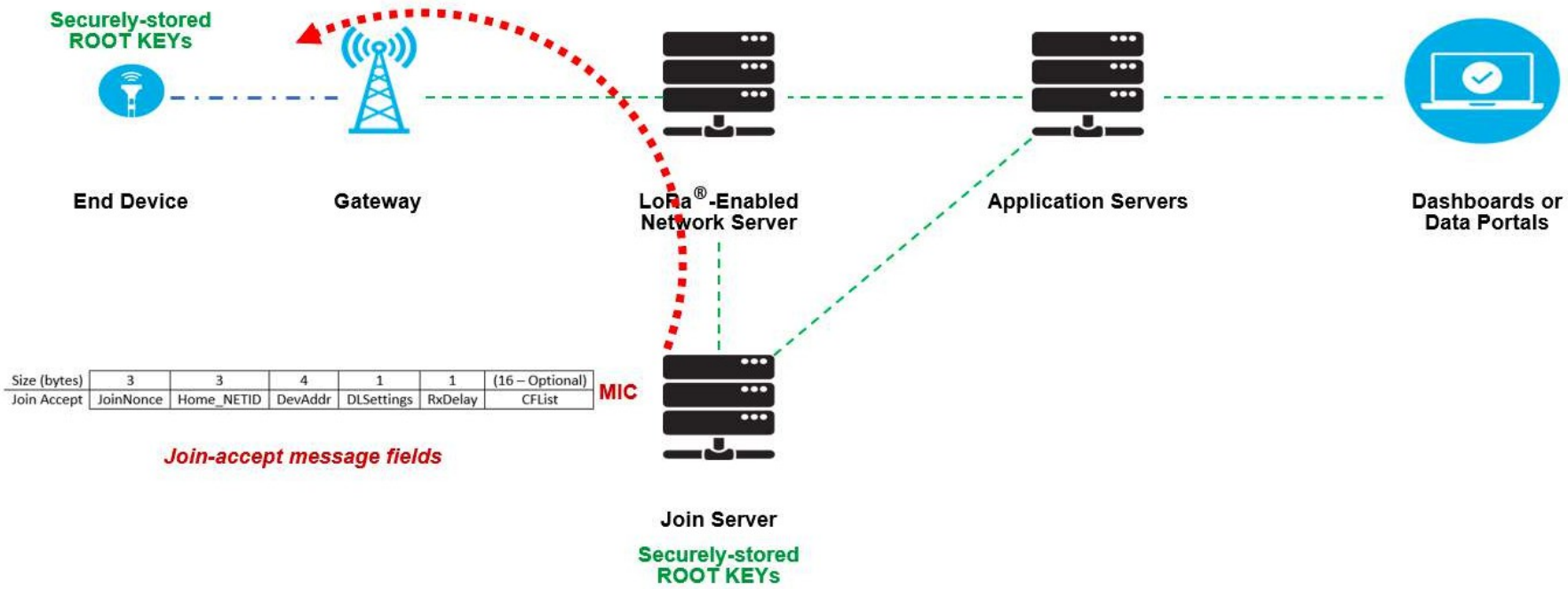
## Join Request




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN Security: OTTA

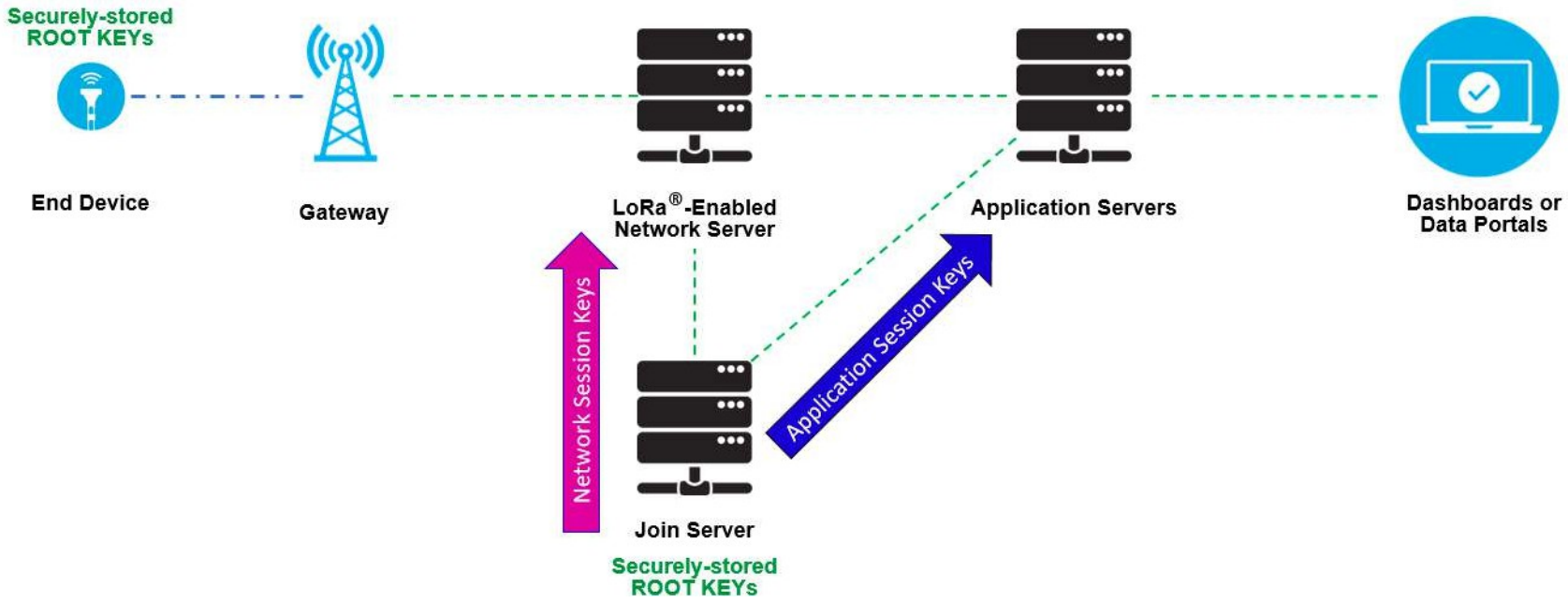
## Join Response




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN Security: OTTA

## Session keys distribution



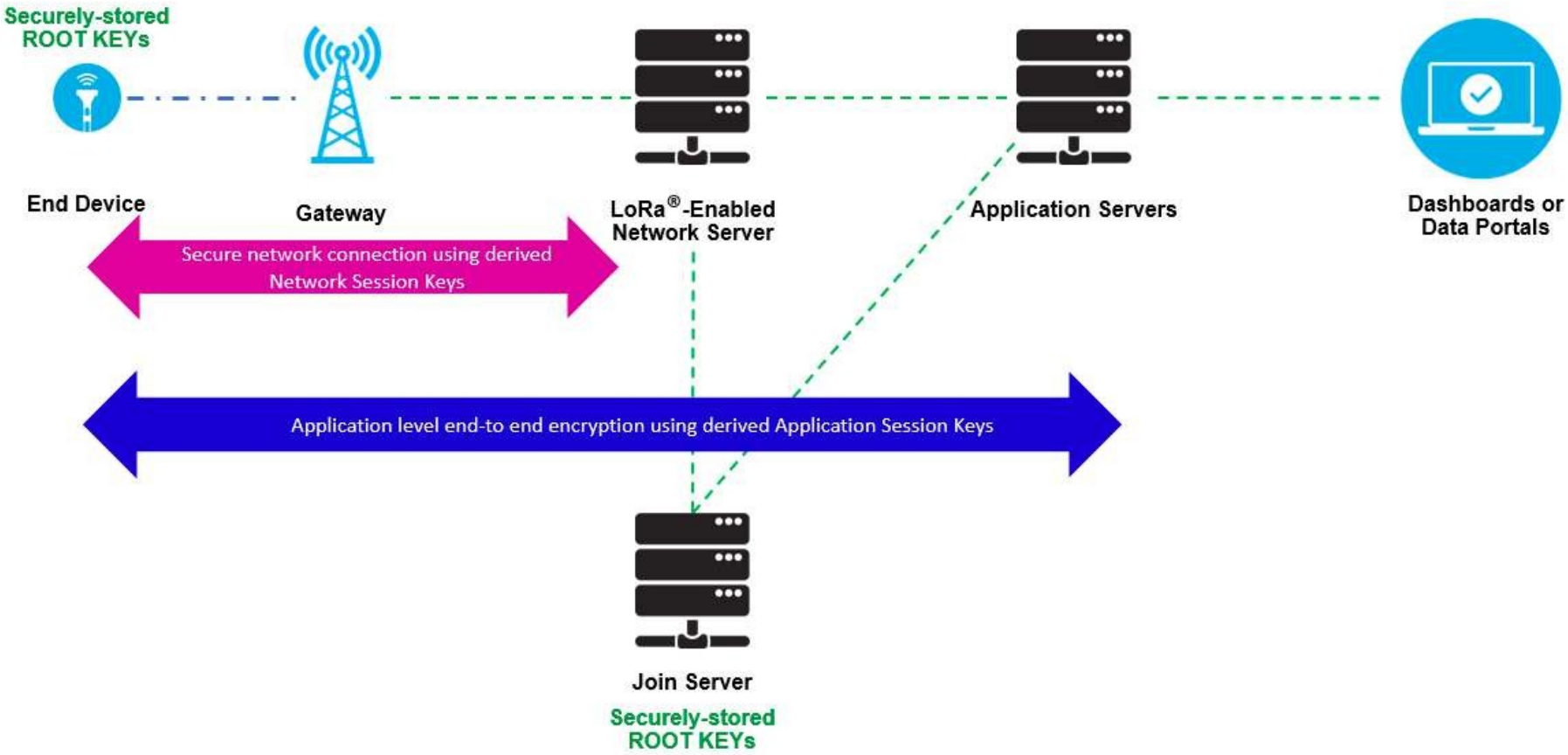

**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# LoRaWAN Security

## LoRaWAN security: Two modes

- **OTAA (Over-the-Air Activation)**
  - Master 128-bit Key (**Appkey**) for initial mutual authentication and the generation of two **session keys**:
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    - Application Session Keys (**AppSKey**) for encryption (confidentiality)
- **ABP (Activation by personalization)**
  - each end-node is provided with **NwkSKey and AppSKey**

# LoRaWAN security: Summary




**source:** https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

# 2- IoT security

# 2.2 Zigbee networks

# Security of some technologies: 802.15.4/Zigbee

- IEEE 802.15.4 defines security features (encryption/integrity algorithm, etc.), MIC size, etc., and makes them available to higher layers.

- Key management is performed at the upper layers (e.g., Zigbee).

- The encryption algorithm used is AES (Advanced Encryption Standard) with a key length of 128 bits.

# 802.15.4 Frame

| Bits: 0–2 | 3 | 4 | 5 | 6 | 7–9 | 10–11 | 12–13 | 14–15 |
|---|---|---|---|---|---|---|---|---|
| Frame Type | Security Enabled | Frame Pending | AR | PAN ID Compression | Reserved | Dest. Addressing Mode | Frame Version | Source Addressing Mode |

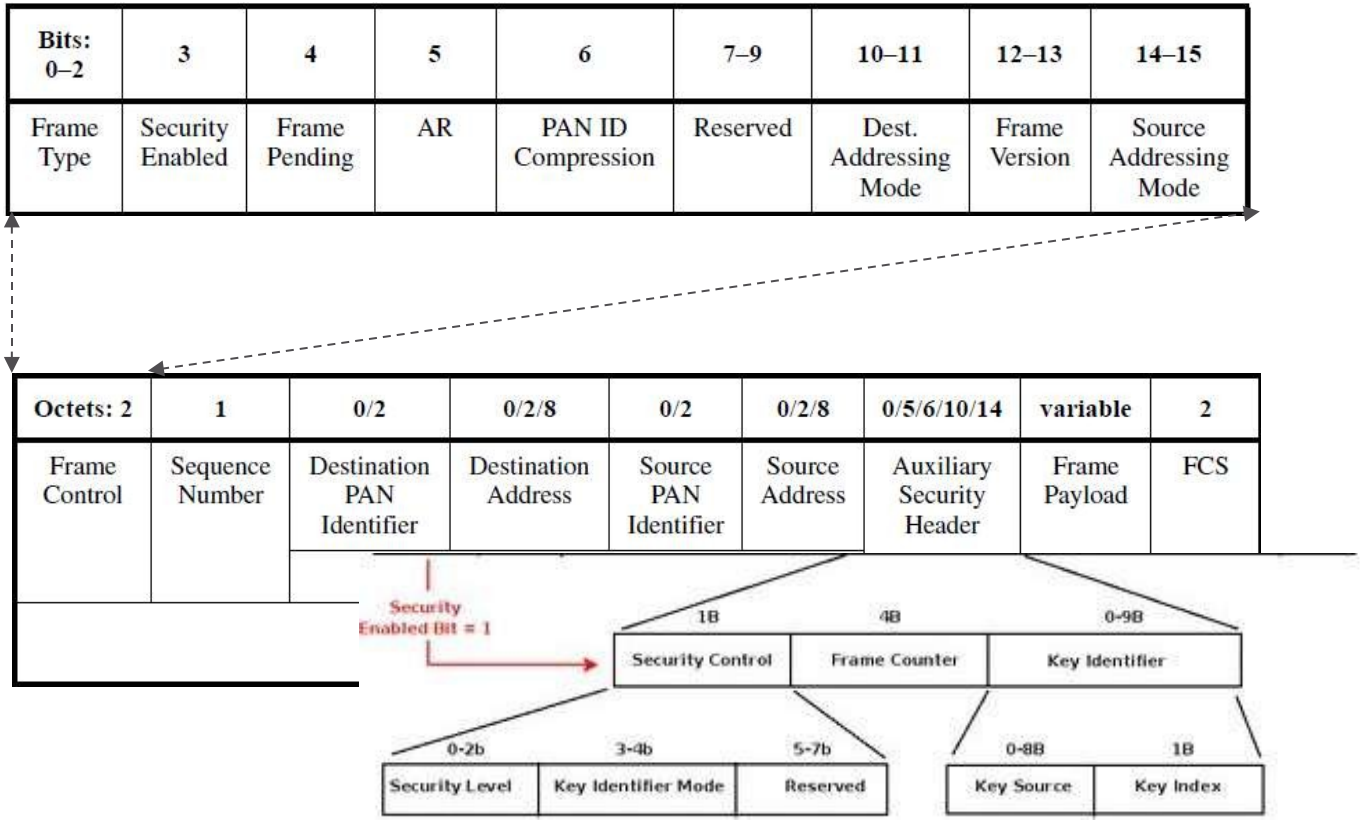
| Octets: 2 | 1 | 0/2 | 0/2/8 | 0/2 | 0/2/8 | 0/5/6/10/14 | variable | 2 |
|---|---|---|---|---|---|---|---|---|
| Frame Control | Sequence Number | Destination PAN Identifier | Destination Address | Source PAN Identifier | Source Address | Auxiliary Security Header | Frame Payload | FCS |
| | | Addressing fields | | | | | | |
| MHR | | | | | | | MAC Payload | MFR |

127 bytes max

Source of figure: Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details: https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

# 802.15.4 Frame



# 802.15.4 Frame: Security level

| | | | | | |
|---|---|---|---|---|---|
| 0x00 | No security | | | Data is not encrypted. Data authenticity is not validated. | |
| 0x01 | AES-CBC-MAC-32 | | MIC-32 | Data is not encrypted. Data authenticity is validated. | |
| 0x02 | AES-CBC-MAC-64 | | MIC-64 | Data is not encrypted. Data authenticity is validated. | |
| 0x03 | AES-CBC-MAC-128 | | MIC-128 | Data is not encrypted. Data authenticity is validated. | |
| 0x04 | AES-CTR | | ENC | Data is encrypted. Data authenticity is not validated. | |
| 0x05 | AES-CCM-32 | | AES-CCM-32 | Data is encrypted. Data authenticity is validated. | |
| 0x06 | AES-CCM-64 | | AES-CCM-64 | Data is encrypted. Data authenticity is validated. | |
| 0x07 | AES-CCM-128 | | AES-CCM-128 | Data is encrypted. Data authenticity is validated. | |

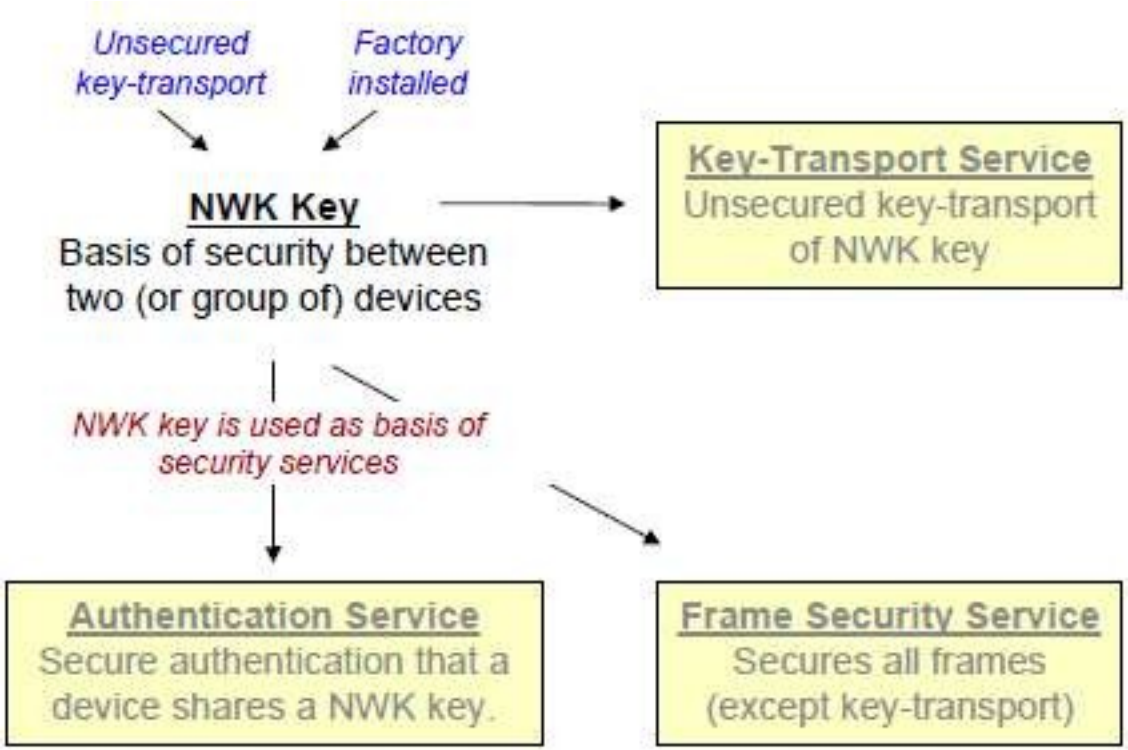
# Security of some technologies: Zigbee

- **Zigbee: security policies**
  - **Residential mode:** Network Key only
  - **Commercial mode**: Master key and Link keys

- **Zigbee Residential mode: (Network key)**
  - A unique 128b key shared among all the devices in the network
  - Pre-installation or Key-transport
  - Each router decrypts and verifies each packet before forwarding it

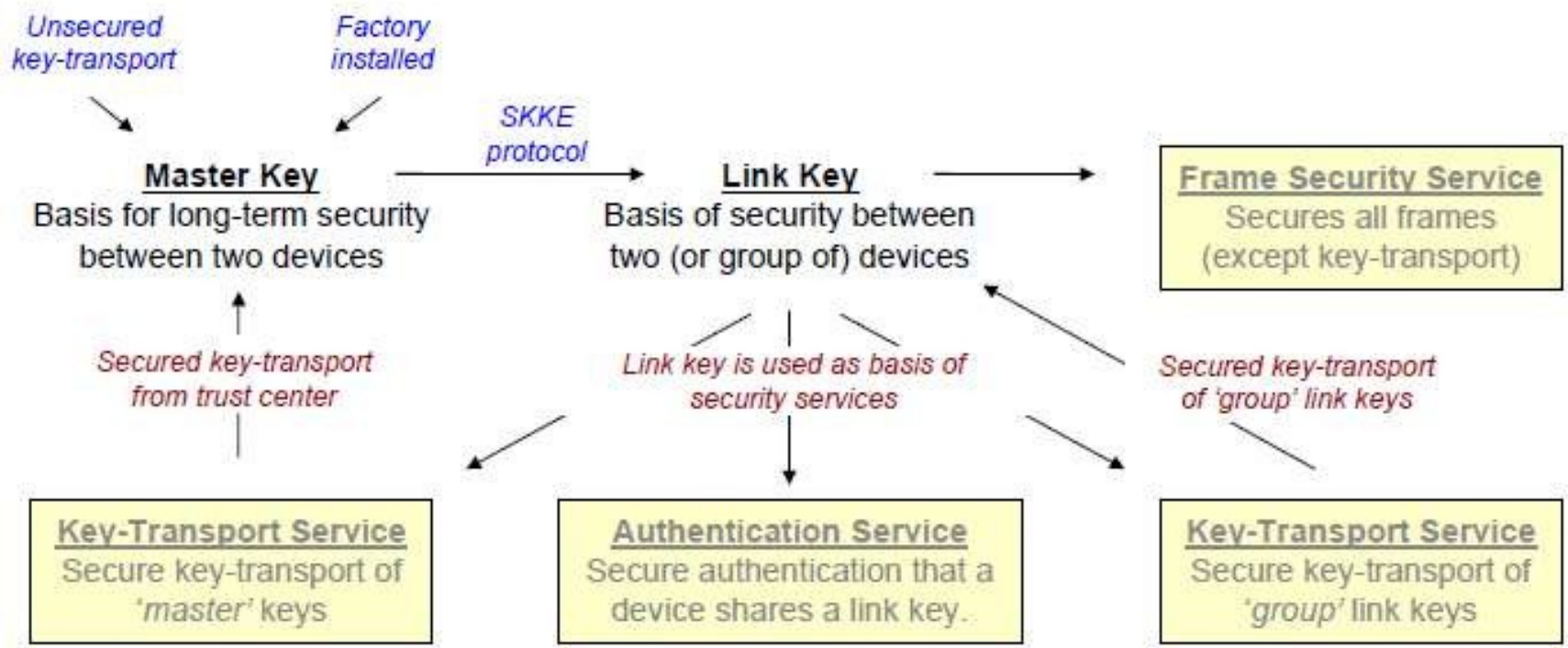
- **Zigbee Commercial mode: (Master and link keys)**
  - Master Keys: pre-installed or Key transport
  - used for Key Establishment Procedure (SKKE) to establish link keys.
  - Link key: unique between each pair of nodes

# Security of some technologies: Zigbee




ZigBee Residential Mode: sourcehttps://sbt-alliance.com/wp-content/uploads/2020/04/Daintree-DT200-GE-Zigbee-Primer-Whitepaper.pdf

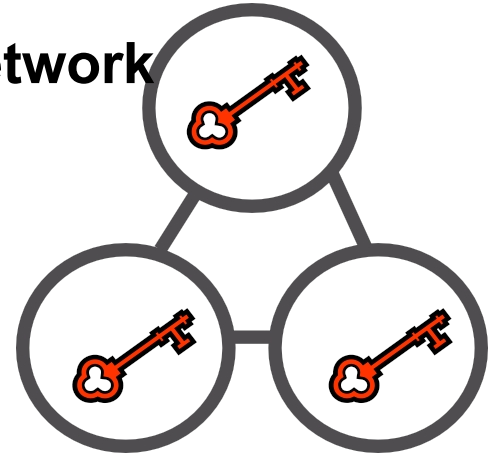
# Security of some technologies: Zigbee



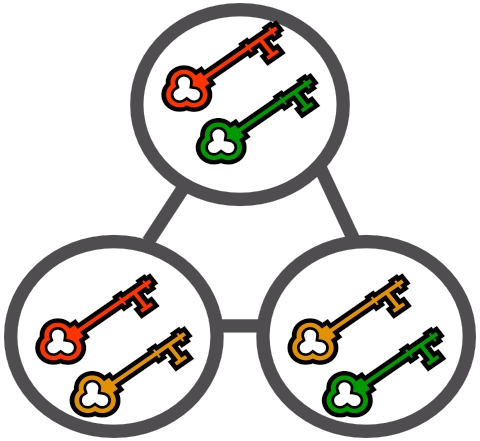

ZigBee Commercial Mode: sourcehttps://sbt-alliance.com/wp-content/uploads/2020/04/Daintree-DT200-GE-Zigbee-Primer-Whitepaper.pdf

# Unique key Vs link keys

- **A unique key to secure the entire network**
  - –Memory efficiency
  - –Very poor network resilience



- **One Key per link**
  - –Perfect resilience
  - –Huge memory requirement



# 2- IoT security

## 2.3 Introduction to research problems

# Key management in Ad-Hoc WSN: Symmetric vs. Asymmetric

**Symmetric cryptosystems**

- Simplicity
- Fast computation
- Small size of required keys
- Symmetric key agreement

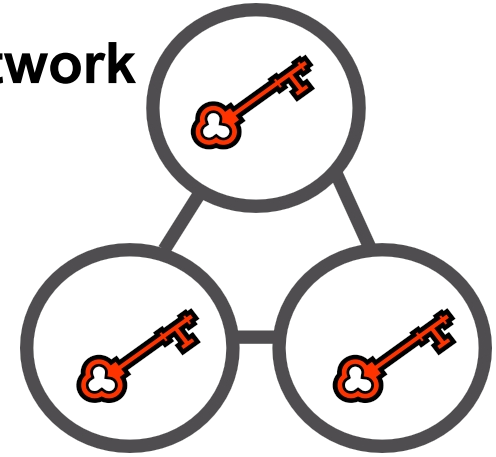
**Asymmetric cryptosystems**

- Efficient key management
- Heavy computations
- Large size keys
- Complex and slower operations

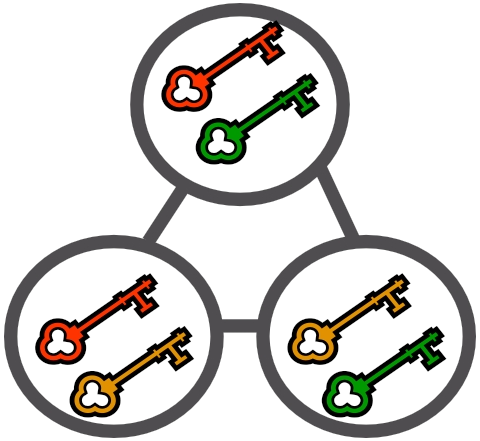
| | | |
|---|---|---|
| Resource limitations (energy, memory and computations) | $\longrightarrow$ | Symmetric key management |
| Lack of infrastructure | $\longrightarrow$ | Key pre-distribution |

# Unique key vs. link keys

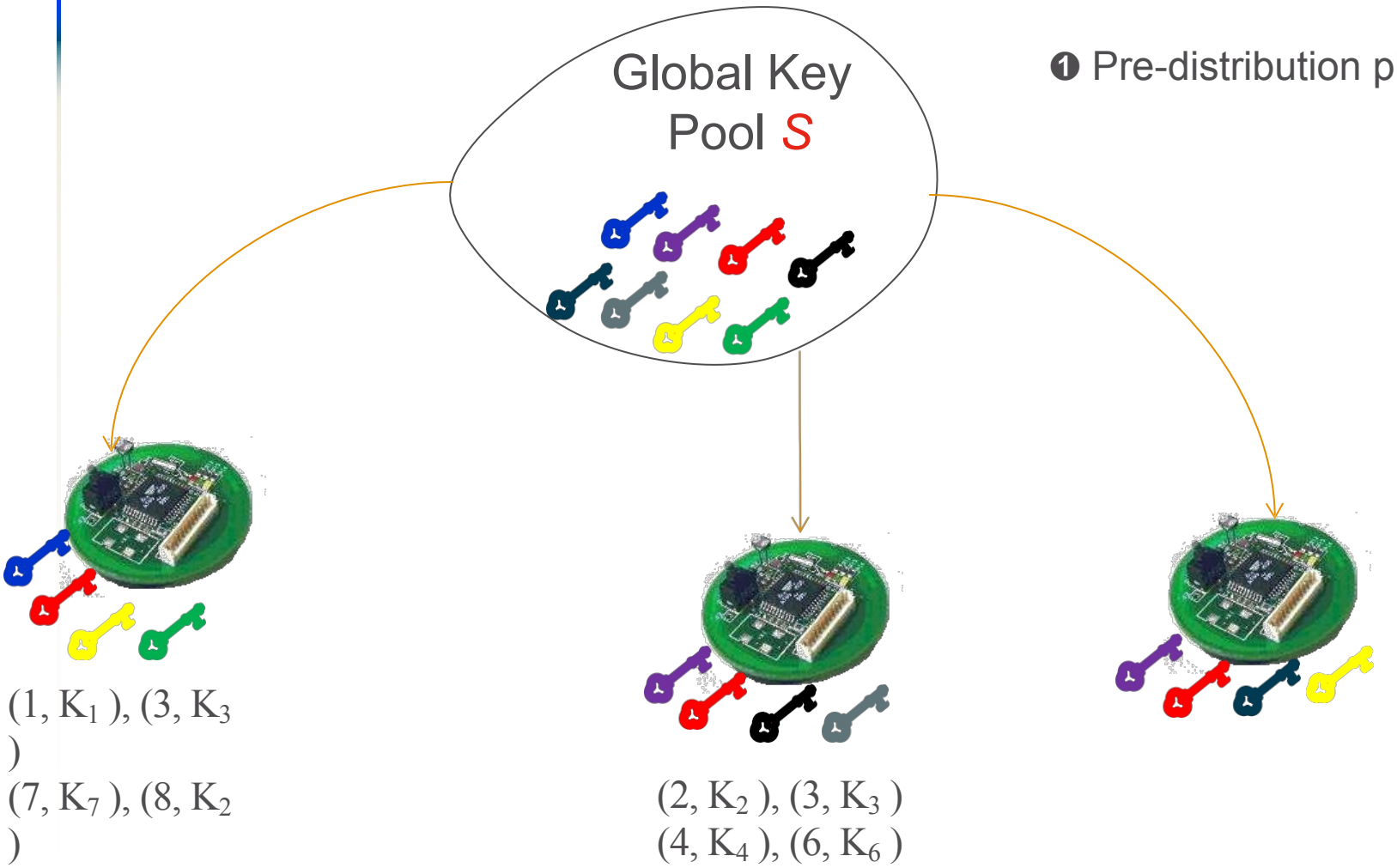
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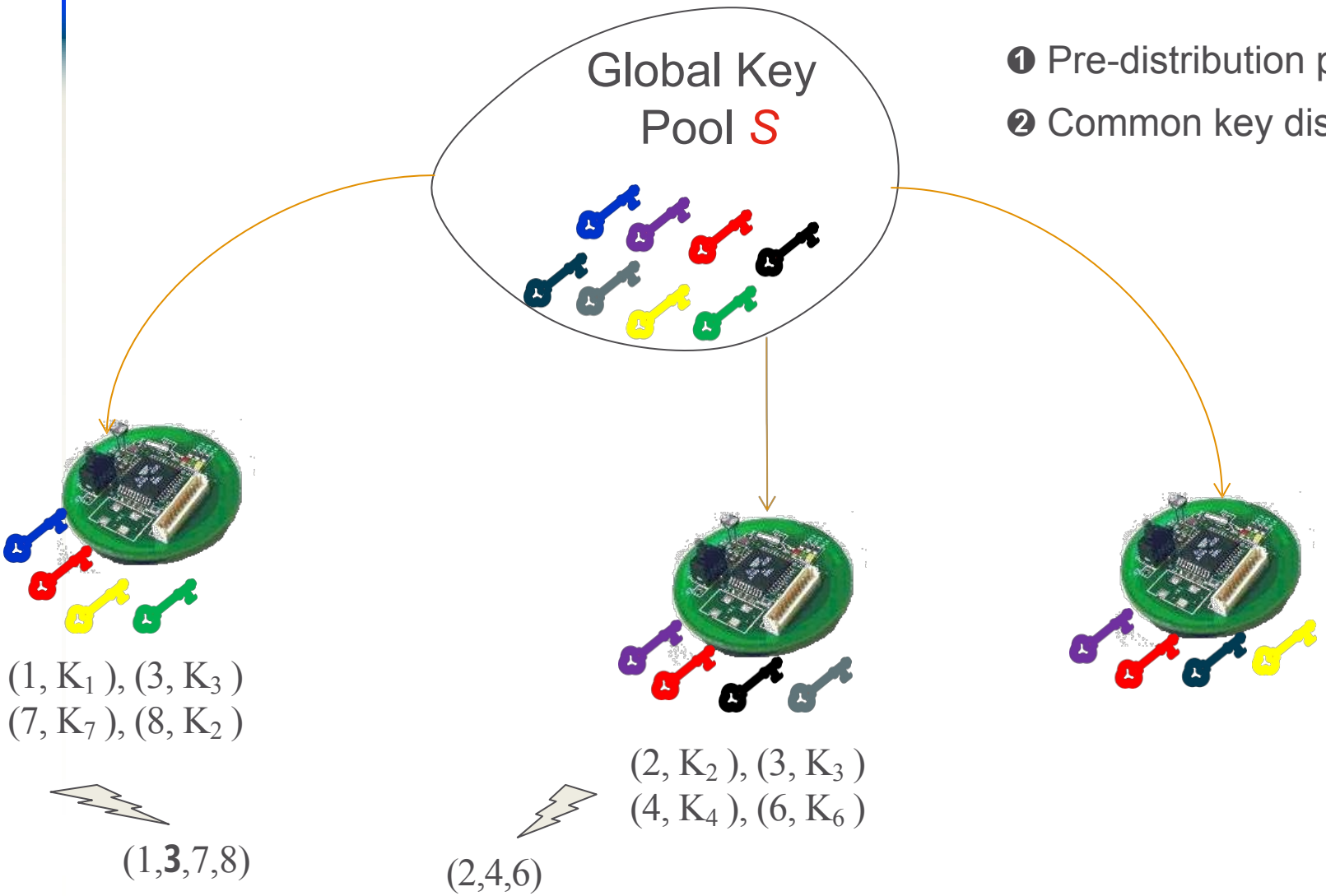
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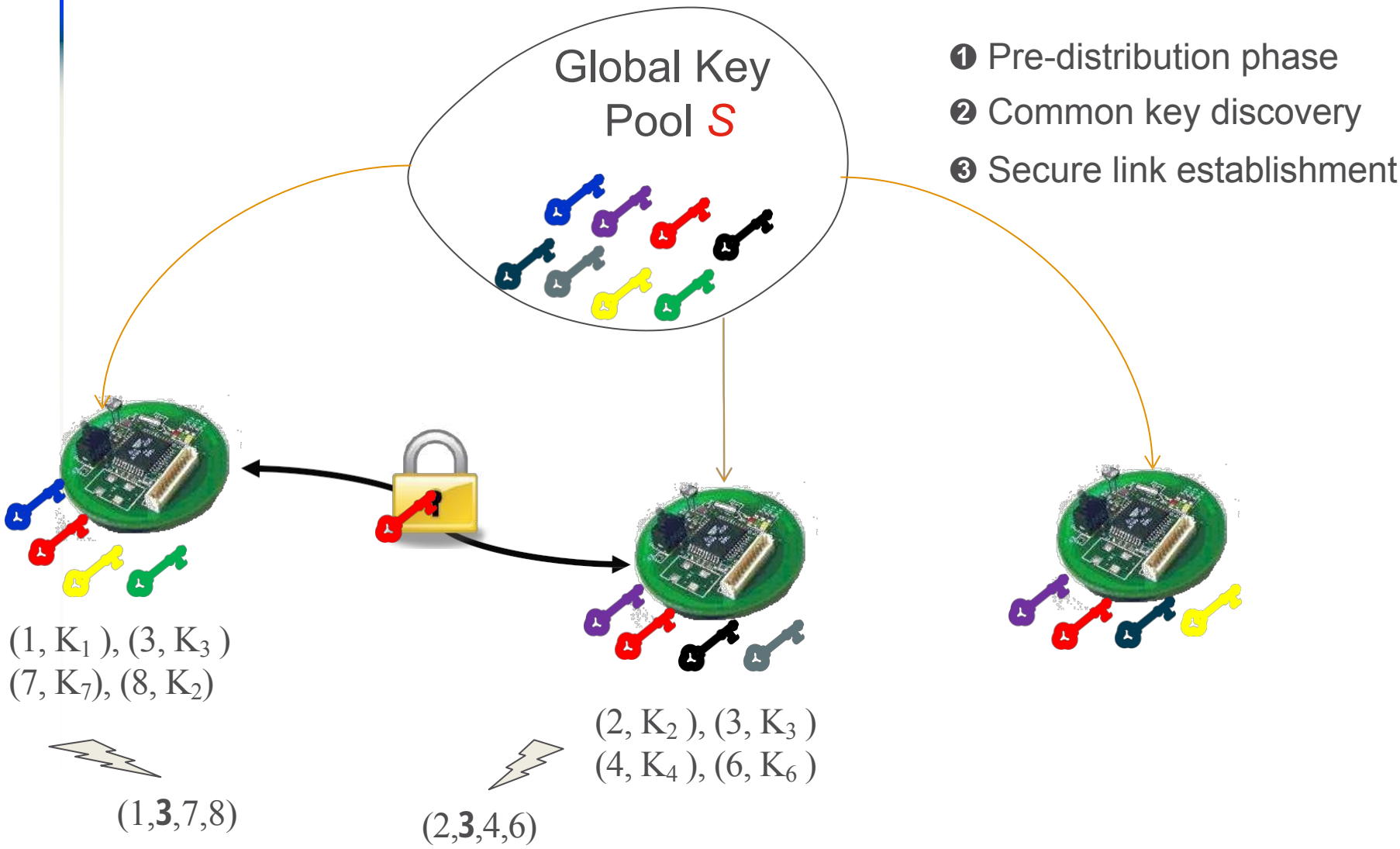
# Key pre-distribution for Ad-Hoc WSN



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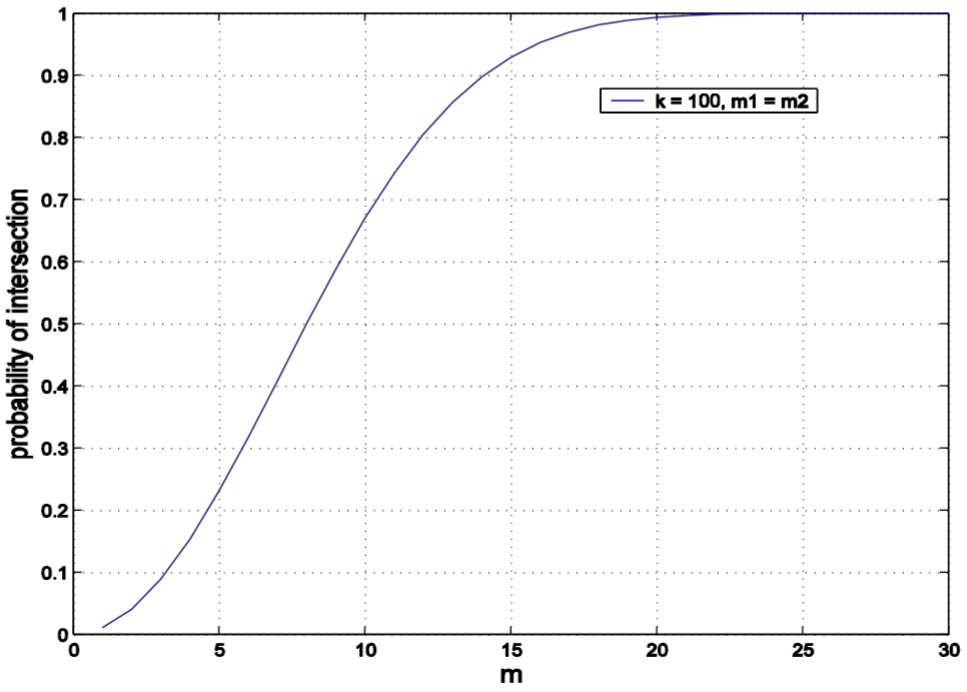
# Evaluation Metrics

- **Key ring size:** Storage overhead
- **Key sharing probability (network secure connectivity):** probabilistic schemes Vs deterministic schemes
- **Network resiliency:** fraction of uncompromised external links when nodes are compromised
- **Network scalability**
- **Communication overhead**
- **Computation overhead**

# Connectivity of the basic scheme

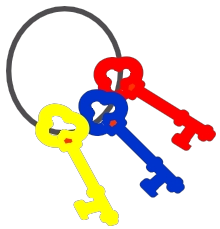
Given a set $S$ of $k$ elements, we randomly choose two subsets $S_1$ and $S_2$ of $m_1$ and $m_2$ elements, respectively, from $S$.

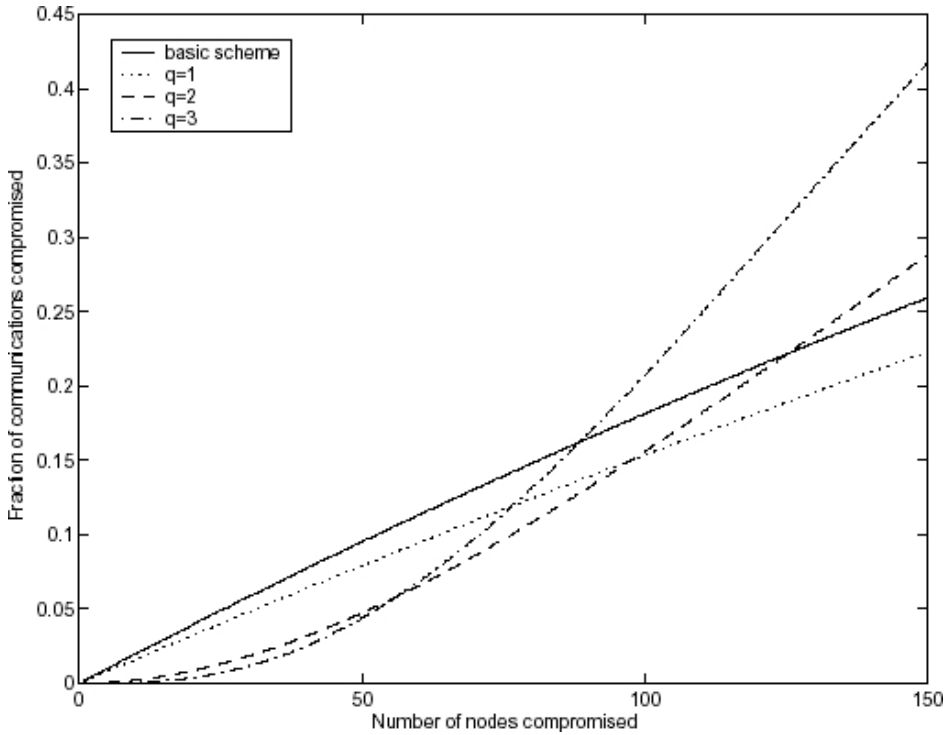
$$\Pr\{S_1 \cap S_2 \neq \emptyset\} = 1 - \frac{(k-m_1)!(k-m_2)!}{k!(k-m_1-m_2)!}$$



# Q-Composite Scheme

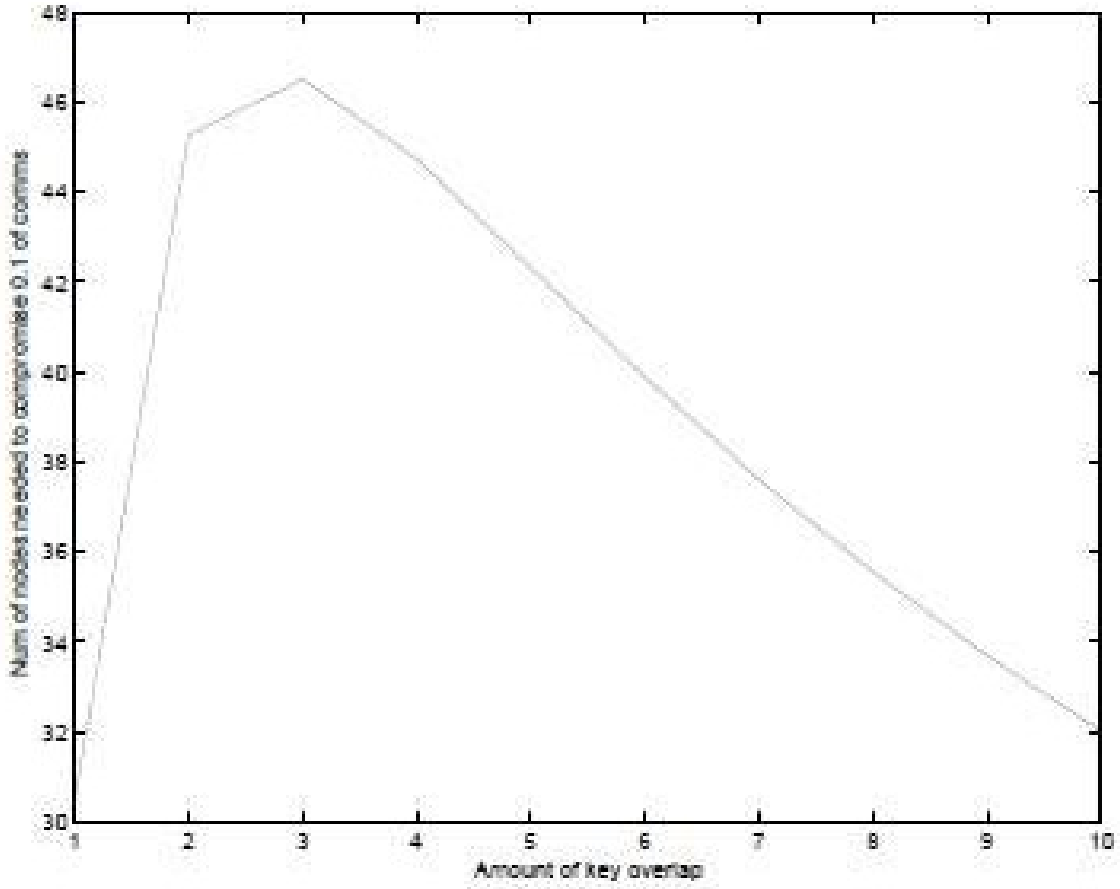
- At least Q common keys to establish a secure link
- The session key is a combination of all common keys






**Fraction of communications compromised k = 200, p = 0.33**

# Q-Composite Scheme




**# nodes needed to compromise 0.1 of comms ( k = 200, p = 0.5)**

# Some key management schemes for WSN

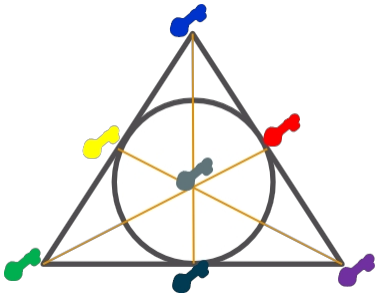
- **Probabilistic schemes**



  – Q-composite scheme

  – Deployment knowledge-based scheme, Polynomial-based schemes, BK scheme, [Ruj et al. 2011] Trade-based scheme.

- **Deterministic schemes**



  –Symmetric Balanced Incomplete Design based scheme. Ex: {1,2,3} {1,4,5} {1,6,7} {2,4,6} {2,5,7} {3, 4, 7} {3, 5, 6}

  – Key pre-distribution using Partially Balanced Incomplete Block Design

# Thank you!